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1 Introduction

This document is a companion reference to the protocol specifications. It describes how **Unicode** strings are compared in Windows protocols and how Windows supports Unicode conversion to earlier **codepages**. For example:

- **UTF-16** string comparison: Provides linguistic-specific comparisons between two Unicode strings and provides the comparison result based on the language and region for a specific user.
- Mapping of UTF-16 strings to earlier ANSI codepages: Converts Unicode strings to strings in the earlier codepages that are used in older versions of Windows and the applications that are written for these earlier codepages.

1.1 Glossary

This document uses the following terms:

code page: An ordered set of characters of a specific script in which a numerical index (code-point value) is associated with each character. Code pages are a means of providing support for character sets and keyboard layouts used in different countries/regions. Devices such as the display and keyboard can be configured to use a specific code page and to switch from one code page (such as the United States) to another (such as Portugal) at the user's request.

double-byte character set (DBCS): A character set that can use more than one byte to represent a single character. A DBCS includes some characters that consist of 1 byte and some characters that consist of 2 bytes. Languages such as Chinese, Japanese, and Korean use DBCS.

IDNA2003: The IDNA2003 specification is defined by a cluster of IETF RFCs: IDNA [\[RFC3490\]](#), Nameprep [\[RFC3491\]](#), Punycode [\[RFC3492\]](#), and Stringprep [\[RFC3454\]](#).

IDNA2008: The IDNA2008 specification is defined by a cluster of IETF RFCs: Internationalized Domain Names for Applications (IDNA): Definitions and Document Framework [\[RFC5890\]](#), Internationalized Domain Names in Applications (IDNA) Protocol [\[RFC5891\]](#), The Unicode Code Points and Internationalized Domain Names for Applications (IDNA) [\[RFC5892\]](#), and Right-to-Left Scripts for Internationalized Domain Names for Applications (IDNA) [\[RFC5893\]](#). There is also an informative document: Internationalized Domain Names for Applications (IDNA): Background, Explanation, and Rationale [\[RFC5894\]](#).

IDNA2008+UTS46: The IDNA2008+UTS46 citation refers to operations that comply with both the and the Unicode IDNA Compatibility Processing [\[TR46\]](#) specifications.

single-byte character set (SBCS): A character encoding in which each character is represented by one byte. Single-byte character sets are limited to 256 characters.

sort key: Numerical representations of a sort element based on locale-specific sorting rules. A sort key consists of several weighted components that represent a character's script, diacritics, case, and additional treatment based on locale.

Unicode: A character encoding standard developed by the Unicode Consortium that represents almost all of the written languages of the world. The **Unicode** standard [\[UNICODE5.0.0/2007\]](#) provides three forms (UTF-8, UTF-16, and UTF-32) and seven schemes (UTF-8, UTF-16, UTF-16 BE, UTF-16 LE, UTF-32, UTF-32 LE, and UTF-32 BE).

UTF-16: A standard for encoding Unicode characters, defined in the Unicode standard, in which the most commonly used characters are defined as double-byte characters. Unless specified otherwise, this term refers to the UTF-16 encoding form specified in [\[UNICODE5.0.0/2007\]](#) section 3.9.

MAY, SHOULD, MUST, SHOULD NOT, MUST NOT: These terms (in all caps) are used as defined in [\[RFC2119\]](#). All statements of optional behavior use either MAY, SHOULD, or SHOULD NOT.

1.2 References

Links to a document in the Microsoft Open Specifications library point to the correct section in the most recently published version of the referenced document. However, because individual documents in the library are not updated at the same time, the section numbers in the documents may not match. You can confirm the correct section numbering by checking the [Errata](#).

1.2.1 Normative References

We conduct frequent surveys of the normative references to assure their continued availability. If you have any issue with finding a normative reference, please contact dochelp@microsoft.com. We will assist you in finding the relevant information.

[CODEPAGEFILES] Microsoft Corporation, "Windows Supported Code Page Data Files.zip", 2009, <https://www.microsoft.com/en-us/download/details.aspx?id=10921>

[ECMA-035] ECMA International, "Character Code Structure and Extension Techniques", 6th edition, ECMA-035, December 1994, <http://www.ecma-international.org/publications/standards/Ecma-035.htm>

[GB18030] Chinese IT Standardization Technical Committee, "Chinese National Standard GB 18030-2005: Information technology - Chinese coded character set", Published in print by the China Standard Press, https://infostore.saiglobal.com/en-us/Standards/SAC-GB-18030-2000-987664_SAIG_SAC_SAC_2308492/

[ISCII] Bureau of Indian Standards, "Indian Script Code for Information Exchange - ISCII", <https://archive.org/details/gov.in.is.13194.1991/page/n4/mode/1up>

[MSDN-SWT] Microsoft Corporation, "Sorting Weight Tables", <http://www.microsoft.com/en-us/download/details.aspx?id=10921>

[MSDN-UCMT/Win8] Microsoft Corporation, "Windows 8 Upper Case Mapping Table", <http://www.microsoft.com/download/en/details.aspx?displaylang=en&id=10921>

[RFC2119] Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, March 1997, <https://www.rfc-editor.org/info/rfc2119>

[RFC2152] Goldsmith, D., and David, M., "UTF-7 A Mail-Safe Transformation Format of Unicode", RFC 2152, May 1997, <https://www.rfc-editor.org/info/rfc2152>

[TR46] Davis, M., and Suignard, M., "Unicode IDNA Compatibility Processing", Unicode Technical Standard #46, September 2012, "", <http://www.unicode.org/reports/tr46/>

[UNICODE-BESTFIT] The Unicode Consortium, "WindowsBestFit", 2006, <http://www.unicode.org/Public/MAPPINGS/VENDORS/MICSFT/WindowsBestFit/>

[UNICODE-COLLATION] The Unicode Consortium, "Unicode Technical Standard #10 Unicode Collation Algorithm", March 2008, <http://www.unicode.org/reports/tr10/>

[UNICODE-README] The Unicode Consortium, "Readme.txt", 2006, <http://unicode.org/Public/MAPPINGS/VENDORS/MICSFT/WindowsBestFit/readme.txt>

[UNICODE5.0.0/CH3] The Unicode Consortium, "Unicode Encoding Forms", 2006, <http://www.unicode.org/versions/Unicode5.0.0/ch03.pdf#G7404>

[UNICODE] The Unicode Consortium, "The Unicode Consortium Home Page", <http://www.unicode.org/>

1.2.2 Informative References

None.

1.3 Overview

This document describes the following protocols when dealing with **Unicode** strings on the Windows platform:

- **UTF-16** string comparison: This string comparison is used to provide a linguistic-specific comparison between two Unicode strings. This scenario provides a string comparison result based on the expectations of users from different languages and different regions.
- The mapping of UTF-16 strings to earlier **codepages**: This scenario is used to convert between Unicode strings and strings in the earlier codepage, which are used by older versions of Windows and applications written for these earlier codepages.

1.4 Applicability Statement

This reference document is applicable as follows:

- To perform **UTF-16** character comparisons in the same manner as Windows. This document only specifies a subset of Windows behaviors that are used by other protocols. It does not document those Windows behaviors that are not used by other protocols.
- To provide the capability to map between UTF-16 strings and earlier **codepages** in the same manner as Windows.

1.5 Standards Assignments

The following standards assignments are used by the Windows Protocols Unicode Reference.

Parameter	Value	Reference
Codepage Data File (section 2.2.2)	Various	[UNICODE-BESTFIT]

2 Messages

The following sections specify how Windows Protocols Unicode Reference messages are transported and Windows Protocols Unicode Reference message syntax.

2.1 Transport

2.2 Message Syntax

2.2.1 Supported Codepage in Windows

Windows assigns an integer, called code page ID, to every supported **codepage**.

Based on the usage, the codepage supported in Windows can be categorized in the following:

- ANSI codepage

Windows codepages are also sometimes referred to as active codepages or system active codepages. Windows always has one currently active Windows codepage. All ANSI Windows functions use the currently active codepage.

The usual ANSI codepage ID for US English is codepage 1252.

Windows codepage 1252, the codepage commonly used for English and other Western European languages, was based on an American National Standards Institute (ANSI) draft. That draft eventually became ISO 8859-1, but Windows codepage 1252 was implemented before the standard became final, and is not exactly the same as ISO 8859-1.

- OEM codepage

- Extended codepage

These codepages cannot be used as ANSI codepages, or OEM codepages. Windows can support conversions between Unicode and these codepages. These codepages are generally used for information exchange purpose with international/national standard or legacy systems. Examples are UTF-8, UTF-7, EBCDIC, and Macintosh codepages.

The following table shows all the supported codepages by Windows. The Codepage ID lists the integer number assigned to a codepage. ANSI/OEM codepages are in bold face. The Codepage Description column describes the codepage. The Codepage notes column lists the category of a codepage and the relevant protocol section in this document to find protocol information.

Codepage ID	Codepage descriptions	Codepage notes
37	IBM EBCDIC US-Canada	Extended codepage; for processing rules, see section 3.1.5.1.1 .
437	OEM United States	OEM codepage; for processing rules, see section 3.1.5.1.1.
500	IBM EBCDIC International	Extended codepage; for processing rules, see section 3.1.5.1.1.
708	Arabic (ASMO 708)	Extended codepage; for processing rules, see section 3.1.5.1.1.
720	Arabic (Transparent ASMO); Arabic (DOS)	Extended codepage; for processing rules, see

Codepage ID	Codepage descriptions	Codepage notes
		section 3.1.5.1.1.
737	OEM Greek (formerly 437G); Greek (DOS)	OEM codepage; for processing rules, see section 3.1.5.1.1.
775	OEM Baltic; Baltic (DOS)	OEM codepage; for processing rules, see section 3.1.5.1.1.
850	OEM Multilingual Latin 1; Western European (DOS)	OEM codepage; for processing rules, see section 3.1.5.1.1.
852	OEM Latin 2; Central European (DOS)	OEM codepage; for processing rules, see section 3.1.5.1.1.
855	OEM Cyrillic (primarily Russian)	OEM codepage; for processing rules, see section 3.1.5.1.1.
857	OEM Turkish; Turkish (DOS)	OEM codepage; for processing rules, see section 3.1.5.1.1.
858	OEM Multilingual Latin 1 + Euro symbol	OEM codepage; for processing rules, see section 3.1.5.1.1.
860	OEM Portuguese; Portuguese (DOS)	OEM codepage; for processing rules, see section 3.1.5.1.1.
861	OEM Icelandic; Icelandic (DOS)	OEM codepage; for processing rules, see section 3.1.5.1.1.
862	OEM Hebrew; Hebrew (DOS)	OEM codepage; for processing rules, see section 3.1.5.1.1.
863	OEM French Canadian; French Canadian (DOS)	OEM codepage; for processing rules, see section 3.1.5.1.1.
864	OEM Arabic; Arabic (864)	OEM codepage; for processing rules, see section 3.1.5.1.1.
865	OEM Nordic; Nordic (DOS)	OEM codepage; for processing rules, see section 3.1.5.1.1.
866	OEM Russian; Cyrillic (DOS)	OEM codepage; for processing rules, see section 3.1.5.1.1.
869	OEM Modern Greek; Greek, Modern (DOS)	OEM codepage; for processing rules, see section 3.1.5.1.1.
870	IBM EBCDIC Multilingual/ROECE (Latin 2); IBM EBCDIC Multilingual Latin 2	Extended codepage; for processing rules, see section 3.1.5.1.1.
874	ANSI/OEM Thai (same as 28605, ISO 8859-15); Thai (Windows)	ANSI codepage; for processing rules, see section 3.1.5.1.1.
875	IBM EBCDIC Greek Modern	Extended codepage; for processing rules, see section 3.1.5.1.1.
932	ANSI/OEM Japanese; Japanese (Shift-JIS)	ANSI/OEM codepage; for processing rules, see section 3.1.5.1.1.
936	ANSI/OEM Simplified Chinese (PRC, Singapore); Chinese Simplified (GB2312)	ANSI/OEM codepage; for processing rules, see section 3.1.5.1.1.

Codepage ID	Codepage descriptions	Codepage notes
949	ANSI/OEM Korean (Unified Hangul Code)	ANSI/OEM codepage; for processing rules, see section 3.1.5.1.1.
950	ANSI/OEM Traditional Chinese (Taiwan; Hong Kong SAR, PRC); Chinese Traditional (Big5)	ANSI/OEM codepage; for processing rules, see section 3.1.5.1.1.
1026	IBM EBCDIC Turkish (Latin 5)	Extended codepage; for processing rules, see section 3.1.5.1.1.
1047	IBM EBCDIC Latin 1/Open System	Extended codepage; for processing rules, see section 3.1.5.1.1.
1140	IBM EBCDIC US-Canada (037 + Euro symbol); IBM EBCDIC (US-Canada-Euro)	Extended codepage; for processing rules, see section 3.1.5.1.1.
1141	IBM EBCDIC Germany (20273 + Euro symbol); IBM EBCDIC (Germany-Euro)	Extended codepage; for processing rules, see section 3.1.5.1.1.
1142	IBM EBCDIC Denmark-Norway (20277 + Euro symbol); IBM EBCDIC (Denmark-Norway-Euro)	Extended codepage; for processing rules, see section 3.1.5.1.1.
1143	IBM EBCDIC Finland-Sweden (20278 + Euro symbol); IBM EBCDIC (Finland-Sweden-Euro)	Extended codepage; for processing rules, see section 3.1.5.1.1.
1144	IBM EBCDIC Italy (20280 + Euro symbol); IBM EBCDIC (Italy-Euro)	Extended codepage; for processing rules, see section 3.1.5.1.1.
1145	IBM EBCDIC Latin America-Spain (20284 + Euro symbol); IBM EBCDIC (Spain-Euro)	Extended codepage; for processing rules, see section 3.1.5.1.1.
1146	IBM EBCDIC United Kingdom (20285 + Euro symbol); IBM EBCDIC (UK-Euro)	Extended codepage; for processing rules, see section 3.1.5.1.1.
1147	IBM EBCDIC France (20297 + Euro symbol); IBM EBCDIC (France-Euro)	Extended codepage; for processing rules, see section 3.1.5.1.1.
1148	IBM EBCDIC International (500 + Euro symbol); IBM EBCDIC (International-Euro)	Extended codepage; for processing rules, see section 3.1.5.1.1.
1149	IBM EBCDIC Icelandic (20871 + Euro symbol); IBM EBCDIC (Icelandic-Euro)	Extended codepage; for processing rules, see section 3.1.5.1.1.
1200	Unicode UTF-16, little-endian byte order (BMP of ISO 10646); available only to managed applications	Not used in Windows.
1201	Unicode UTF-16, big-endian byte order; available only to managed applications	Not used in Windows.
1250	ANSI Central European; Central European (Windows)	ANSI codepage; for processing rules, see section 3.1.5.1.1.
1251	ANSI Cyrillic; Cyrillic (Windows)	ANSI codepage; for processing rules, see section 3.1.5.1.1.
1252	ANSI Latin 1; Western European (Windows)	ANSI codepage; for processing rules, see section 3.1.5.1.1.
1253	ANSI Greek; Greek (Windows)	ANSI codepage; for processing rules, see section

Codepage ID	Codepage descriptions	Codepage notes
		3.1.5.1.1.
1254	ANSI Turkish; Turkish (Windows)	ANSI codepage; for processing rules, see section 3.1.5.1.1.
1255	ANSI Hebrew; Hebrew (Windows)	ANSI codepage; for processing rules, see section 3.1.5.1.1.
1256	ANSI Arabic; Arabic (Windows)	ANSI codepage; for processing rules, see section 3.1.5.1.1.
1257	ANSI Baltic; Baltic (Windows)	ANSI codepage; for processing rules, see section 3.1.5.1.1.
1258	ANSI/OEM Vietnamese; Vietnamese (Windows)	ANSI codepage; for processing rules, see section 3.1.5.1.1.
1361	Korean (Johab)	Extended codepage; for processing rules, see section 3.1.5.1.1.
10000	MAC Roman; Western European (Mac)	Extended codepage; for processing rules, see section 3.1.5.1.1.
10001	Japanese (Mac)	Extended codepage; for processing rules, see section 3.1.5.1.1.
10002	MAC Traditional Chinese (Big5); Chinese Traditional (Mac)	Extended codepage; for processing rules, see section 3.1.5.1.1.
10003	Korean (Mac)	Extended codepage; for processing rules, see section 3.1.5.1.1.
10004	Arabic (Mac)	Extended codepage; for processing rules, see section 3.1.5.1.1.
10005	Hebrew (Mac)	Extended codepage; for processing rules, see section 3.1.5.1.1.
10006	Greek (Mac)	Extended codepage; for processing rules, see section 3.1.5.1.1.
10007	Cyrillic (Mac)	Extended codepage; for processing rules, see section 3.1.5.1.1.
10008	MAC Simplified Chinese (GB 2312); Chinese Simplified (Mac)	Extended codepage; for processing rules, see section 3.1.5.1.1.
10010	Romanian (Mac)	Extended codepage; for processing rules, see section 3.1.5.1.1.
10017	Ukrainian (Mac)	Extended codepage; for processing rules, see section 3.1.5.1.1.
10021	Thai (Mac)	Extended codepage; for processing rules, see section 3.1.5.1.1.
10029	MAC Latin 2; Central European (Mac)	Extended codepage; for processing rules, see section 3.1.5.1.1.
10079	Icelandic (Mac)	Extended codepage; for processing rules, see section 3.1.5.1.1.

Codepage ID	Codepage descriptions	Codepage notes
10081	Turkish (Mac)	Extended codepage; for processing rules, see section 3.1.5.1.1.
10082	Croatian (Mac)	Extended codepage; for processing rules, see section 3.1.5.1.1.
12000	Unicode UTF-32, little-endian byte order; available only to managed applications	Not used in Windows.
12001	Unicode UTF-32, big-endian byte order; available only to managed applications	Not used in Windows.
20000	CNS Taiwan; Chinese Traditional (CNS)	Extended codepage; for processing rules, see section 3.1.5.1.1.
20001	TCA Taiwan	Extended codepage; for processing rules, see section 3.1.5.1.1.
20002	Eten Taiwan; Chinese Traditional (Eten)	Extended codepage; for processing rules, see section 3.1.5.1.1.
20003	IBM5550 Taiwan	Extended codepage; for processing rules, see section 3.1.5.1.1.
20004	TeleText Taiwan	Extended codepage; for processing rules, see section 3.1.5.1.1.
20005	Wang Taiwan	Extended codepage; for processing rules, see section 3.1.5.1.1.
20105	IA5 (IRV International Alphabet No. 5, 7-bit); Western European (IA5)	Extended codepage; for processing rules, see section 3.1.5.1.1.
20106	IA5 German (7-bit)	Extended codepage; for processing rules, see section 3.1.5.1.1.
20107	IA5 Swedish (7-bit)	Extended codepage; for processing rules, see section 3.1.5.1.1.
20108	IA5 Norwegian (7-bit)	Extended codepage; for processing rules, see section 3.1.5.1.1.
20127	US-ASCII (7-bit)	Extended codepage; for processing rules, see section 3.1.5.1.1.
20261	T.61	Extended codepage; for processing rules, see section 3.1.5.1.1.
20269	ISO 6937 Non-Spacing Accent	Extended codepage; for processing rules, see section 3.1.5.1.1.
20273	IBM EBCDIC Germany	Extended codepage; for processing rules, see section 3.1.5.1.1.
20277	IBM EBCDIC Denmark-Norway	Extended codepage; for processing rules, see section 3.1.5.1.1.
20278	IBM EBCDIC Finland-Sweden	Extended codepage; for processing rules, see section 3.1.5.1.1.
20280	IBM EBCDIC Italy	Extended codepage; for processing rules, see section 3.1.5.1.1.

Codepage ID	Codepage descriptions	Codepage notes
20284	IBM EBCDIC Latin America-Spain	Extended codepage; for processing rules, see section 3.1.5.1.1.
20285	IBM EBCDIC United Kingdom	Extended codepage; for processing rules, see section 3.1.5.1.1.
20290	IBM EBCDIC Japanese Katakana Extended	Extended codepage; for processing rules, see section 3.1.5.1.1.
20297	IBM EBCDIC France	Extended codepage; for processing rules, see section 3.1.5.1.1.
20420	IBM EBCDIC Arabic	Extended codepage; for processing rules, see section 3.1.5.1.1.
20423	IBM EBCDIC Greek	Extended codepage; for processing rules, see section 3.1.5.1.1.
20424	IBM EBCDIC Hebrew	Extended codepage; for processing rules, see section 3.1.5.1.1.
20833	IBM EBCDIC Korean Extended	Extended codepage; for processing rules, see section 3.1.5.1.1.
20838	IBM EBCDIC Thai	Extended codepage; for processing rules, see section 3.1.5.1.1.
20866	Russian (KOI8-R); Cyrillic (KOI8-R)	Extended codepage; for processing rules, see section 3.1.5.1.1.
20871	IBM EBCDIC Icelandic	Extended codepage; for processing rules, see section 3.1.5.1.1.
20880	IBM EBCDIC Cyrillic Russian	Extended codepage; for processing rules, see section 3.1.5.1.1.
20905	IBM EBCDIC Turkish	Extended codepage; for processing rules, see section 3.1.5.1.1.
20924	IBM EBCDIC Latin 1/Open System (1047 + Euro symbol)	Extended codepage; for processing rules, see section 3.1.5.1.1.
20932	Japanese (JIS 0208-1990 and 0121-1990)	Extended codepage; for processing rules, see section 3.1.5.1.1.
20936	Simplified Chinese (GB2312); Chinese Simplified (GB2312-80)	Extended codepage; for processing rules, see section 3.1.5.1.1.
20949	Korean Wansung	Extended codepage; for processing rules, see section 3.1.5.1.1.
21025	IBM EBCDIC Cyrillic Serbian-Bulgarian	Extended codepage; for processing rules, see section 3.1.5.1.1.
21027	Ext Alpha Lowercase	Extended codepage; for processing rules, see section 3.1.5.1.1. NOTE: Although this codepage is supported, it has no known use.
21866	Ukrainian (KOI8-U); Cyrillic (KOI8-U)	Extended codepage; for processing rules, see section 3.1.5.1.1.
28591	ISO 8859-1 Latin 1; Western European	Extended codepage; for processing rules, see

Codepage ID	Codepage descriptions	Codepage notes
	(ISO)	section 3.1.5.1.1.
28592	ISO 8859-2 Central European; Central European (ISO)	Extended codepage; for processing rules, see section 3.1.5.1.1.
28593	ISO 8859-3 Latin 3	Extended codepage; for processing rules, see section 3.1.5.1.1.
28594	ISO 8859-4 Baltic	Extended codepage; for processing rules, see section 3.1.5.1.1.
28595	ISO 8859-5 Cyrillic	Extended codepage; for processing rules, see section 3.1.5.1.1.
28596	ISO 8859-6 Arabic	Extended codepage; for processing rules, see section 3.1.5.1.1.
28597	ISO 8859-7 Greek	Extended codepage; for processing rules, see section 3.1.5.1.1.
28598	ISO 8859-8 Hebrew; Hebrew (ISO-Visual)	Extended codepage; for processing rules, see section 3.1.5.1.1.
28599	ISO 8859-9 Turkish	Extended codepage; for processing rules, see section 3.1.5.1.1.
28603	ISO 8859-13 Estonian	Extended codepage; for processing rules, see section 3.1.5.1.1.
28605	ISO 8859-15 Latin 9	Extended codepage; for processing rules, see section 3.1.5.1.1.
38598	ISO 8859-8 Hebrew; Hebrew (ISO-Logical)	Extended codepage; for processing rules, see section 3.1.5.1.1. Use [CODEPAGEFILES] 28598.txt.
50220	ISO 2022 Japanese with no halfwidth Katakana; Japanese (JIS)	Extended codepage; for processing rules, see section 3.1.5.1.1.
50221	ISO 2022 Japanese with halfwidth Katakana; Japanese (JIS-Allow 1 byte Kana)	Extended codepage; for processing rules, see section 3.1.5.1.2 .
50222	ISO 2022 Japanese JIS X 0201-1989; Japanese (JIS-Allow 1 byte Kana - SO/SI)	Extended codepage; for processing rules, see section 3.1.5.1.2.
50225	ISO 2022 Korean	Extended codepage; for processing rules, see section 3.1.5.1.2.
50227	ISO 2022 Simplified Chinese; Chinese Simplified (ISO 2022)	Extended codepage; for processing rules, see section 3.1.5.1.2.
50229	ISO 2022 Traditional Chinese	Extended codepage; for processing rules, see section 3.1.5.1.2.
51949	EUC Korean	Extended codepage; for processing rules, see section 3.1.5.1.2. Use [CODEPAGEFILES] 20949.txt.
52936	HZ-GB2312 Simplified Chinese; Chinese Simplified (HZ)	Extended codepage; for processing rules, see section 3.1.5.1.2.
54936	GB18030 Simplified Chinese (4 byte); Chinese Simplified (GB18030)	Extended codepage; for processing rules, see section 3.1.5.1.3 .

Codepage ID	Codepage descriptions	Codepage notes
57002	ISCII Devanagari	Extended codepage; for processing rules, see section 3.1.5.1.4 .
57003	ISCII Bengali	Extended codepage; for processing rules, see section 3.1.5.1.4.
57004	ISCII Tamil	Extended codepage; for processing rules, see section 3.1.5.1.4.
57005	ISCII Telugu	Extended codepage; for processing rules, see section 3.1.5.1.4.
57006	ISCII Assamese	Extended codepage; for processing rules, see section 3.1.5.1.4.
57007	ISCII Odia (was Oriya)	Extended codepage; for processing rules, see section 3.1.5.1.4.
57008	ISCII Kannada	Extended codepage; for processing rules, see section 3.1.5.1.4.
57009	ISCII Malayalam	Extended codepage; for processing rules, see section 3.1.5.1.4.
57010	ISCII Gujarati	Extended codepage; for processing rules, see section 3.1.5.1.4.
57011	ISCII Punjabi	Extended codepage; for processing rules, see section 3.1.5.1.4.
65000	Unicode (UTF-7)	Extended codepage; for processing rules, see section 3.1.5.1.5 .
65001	Unicode (UTF-8)	Extended codepage; for processing rules, see section 3.1.5.1.6 .

2.2.2 Supported Codepage Data Files

The mapping of **UTF-16** strings to **codepages** relies on codepage data files to provide conversion data. These codepage data files map **Unicode** characters to characters in a **single-byte character set (SBCS)** or **double-byte character set (DBCS)**.

The data files of supported system codepages are published as specified in [\[CODEPAGEFILES\]](#), [\[UNICODE\]](#), and [\[UNICODE-BESTFIT\]](#). The location identification uses a simple file-naming convention, which is bestfitxxxx.txt, where xxxx is the codepage number. For example, bestfit950.txt contains the data for codepage 950, and bestfit1252.txt contains the data for codepage 1252.

The pseudocode assumes all these codepage files are available.

2.2.2.1 Codepage Data File Format

The Readme.txt (as specified in [\[UNICODE-README\]](#)) provides details about the **codepages** files and the file format. This section specifies information about the pseudocode of mapping **UTF-16** strings to earlier codepages by taking the content from the Readme.txt.

Each file has sections of keyword tags and records. Any text after ";" is ignored as blank lines. Fields are delimited by one or more space or tab characters. Each section begins with one of the following tags:

- CODEPAGE ([UNICODE-README])
- CPINFO ([UNICODE-README])
- [MBTABLE \(section 2.2.2.1.2\)](#)
- [WCTABLE \(section 2.2.2.1.1\)](#)
- [DBCSTRANGE \(section 2.2.2.1.3\)](#) (DBCS codepages only)
- DBCSTABLE (section 2.2.2.1.3) (DBCS codepages only)

2.2.2.1.1 WCTABLE

The WCTABLE tag marks the start of the mapping from Unicode **UTF-16** to MultiByte bytes. It has one field.

Field 1: The number of records of **Unicode** to byte mappings. Note that this field is often more than the number of roundtrip mappings that are supported by the **codepage** due to Windows best-fit behavior.

An example of the WCTABLE tag is:

```
WCTABLE 698
```

The Unicode UTF-16 mapping records follow the WCTABLE section. These mapping records are in two forms: **single-byte** or **double-byte** codepages. Both forms have two fields.

Field 1: The Unicode UTF-16 code point for the character being converted.

Field 2: The single byte that this UTF-16 code point maps to. This can be a best-fit mapping.

The following example shows Unicode to byte-mapping records for SBCSs.

```
0x0000 0x00; Null
0x0001 0x01; Start Of Heading
...
0x0061 0x61; Latin Small Letter A
0x0062 0x62; Latin Small Letter B
0x0063 0x63; Latin Small Letter C
...
0x221e 0x38; Infinity << Best Fit Mapping
...
0xff41 0x61; Fullwidth Latin Small Letter A << Best Fit Mapping
0xff42 0x62; Fullwidth Latin Small Letter B << Best Fit Mapping
0xff43 0x63; Fullwidth Latin Small Letter C << Best Fit Mapping
...
```

Field 1: The Unicode UTF-16 code point for the character being converted.

Field 2: The byte or bytes that this code point maps to as a 16-bit value. The high byte is the lead byte, and the low byte is the trail byte. If the high byte is 0, this is a single-byte code point with the value of the low byte and no lead byte is emitted.

The following example shows Unicode to byte-mapping records for DBCSs.

```

0x0000 0x0000; Null
0x0001 0x0001; Start Of Heading
...
0x0061 0x0061; a
0x0062 0x0062; b
0x0063 0x0063; c
...
0x221e 0x8187; Infinity
...
0xff41 0x8281; Fullwidth a
0xff42 0x8282; Fullwidth b
0xff43 0x8283; Fullwidth c
...

```

2.2.2.1.2 MBTABLE

The MBTABLE tag marks the start of the mapping from single-byte bytes to **Unicode UTF-16**. It has one field.

Field 1: The number of records of single-byte to Unicode mappings.

An example of the MBTABLE tag is:

```
MBTABLE 196
```

The Unicode UTF-16 mapping records follow the MBTABLE section. These mapping records have two fields.

Field 1: The single byte character of the codepage.

Field 2: The Unicode UTF-16 code point that the **codepage** character maps to.

The following example shows mapping records for codepage 932.

```

0x00 0x0000; Null
0x01 0x0001; Start Of Heading
0x02 0x0002; Start Of Text
0x03 0x0003; End Of Text
0x04 0x0004; End Of Transmission
0x05 0x0005; Enquiry
0x06 0x0006; Acknowledge
0x07 0x0007; Bell
0x08 0x0008; Backspace
...
0xa1 0xff61; Halfwidth Ideographic Period
0xa2 0xff62; Halfwidth Opening Corner Bracket
0xa3 0xff63; Halfwidth Closing Corner Bracket
0xa4 0xff64; Halfwidth Ideographic Comma
0xa5 0xff65; Halfwidth Katakana Middle Dot
0xa6 0xff66; Halfwidth Katakana Wo
0xa7 0xff67; Halfwidth Katakana Small A
0xa8 0xff68; Halfwidth Katakana Small I
0xa9 0xff69; Halfwidth Katakana Small U
0xaa 0xff6a; Halfwidth Katakana Small E
0xab 0xff6b; Halfwidth Katakana Small O
0xac 0xff6c; Halfwidth Katakana Small Ya

```

2.2.2.1.3 DBCSRANGE

The DBCSRANGE tag marks the start of the mapping from double-byte bytes to **Unicode UTF-16**. It has one field.

Field 1: The number of records of lead byte ranges.

An example of the DBCSRANGE tag is:

```
DBCSRANGE 2
```

The Lead Byte Range records follow the DBCSRANGE section. These mapping records have two fields.

Field 1: The start of lead byte range.

Field 2: The end of lead byte range.

The following example shows one of the Lead Byte Range records for **codepage** 932. In this codepage, it has one range of lead byte, starting from 0x81 (decimal 129) to 0x9f (decimal 159). So there are 31 lead bytes in this example (159 – 129 + 1). Each lead byte will have a corresponding DBCSRANGE.

```
0x81 0x9f; Lead Byte Range
```

A group of DBCSTABLE sections follows the lead-byte range record. Each lead byte will have a corresponding DBCSTABLE section. In each DBCSTABLE section, it has one field.

Field 1: This field is the number of trail byte mappings for the lead byte.

The lead byte of the first DBCSTABLE is the first lead byte of the previous Lead Byte Range record. Each subsequent DBCSTABLE is for the next consecutive lead byte value.

The following example shows the first DBCSTABLE for codepage 932. This is for lead byte 0x81.

```
DBCSTABLE 147; LeadByte = 0x81
```

The DBCSTABLE record describes the mappings available for a particular lead byte. The comment is ignored but descriptive.

Field 1: This field is the trail byte to map from.

Field 2: This field is the Unicode UTF-16 code point that this lead byte/trail byte combination map to.

The following example shows DBCSTABLE records for codepage 932 for lead byte 0x81.

```
0x40 0x3000; Ideographic Space  
0x41 0x3001; Ideographic Comma  
...
```

3 Protocol Details

The following sections specify details of the Windows Protocols Unicode Reference, including abstract data models and message processing rules.

3.1 Client Details

3.1.1 Abstract Data Model

This section describes a conceptual model of possible data organization that an implementation maintains to participate in this protocol. The described organization is provided to facilitate the explanation of how the protocol behaves. This document does not mandate that implementations adhere to this model as long as their external behavior is consistent with what is described in this document.

No abstract data model is needed.

3.1.2 Timers

None.

3.1.3 Initialization

None.

3.1.4 Higher-Layer Triggered Events

None.

3.1.5 Message Processing Events and Sequencing Rules

3.1.5.1 Mapping Between UTF-16 Strings and Legacy Codepages

3.1.5.1.1 Mapping Between UTF-16 Strings and Legacy Codepages Using CodePage Data File

This process maps between a Unicode string that is encoded in UTF-16 and a string in a specified codepage by using a codepage data file specified in [2.2.2.1](#).

3.1.5.1.1.1 Pseudocode for Accessing a Record in the Codepage Data File

This section contains the pseudocode that is used to read information from the **codepage** file. The following example is taken from codepage data file 950.txt.

OPEN SECTION indicates that queries for records in a specific section are made. To open the following section with the WCTABLE label, the following syntax is used. The OPEN SECTION is accessible by using the WideCharMapping name.

```
OPEN SECTION WideCharMapping
    where section name is WCTABLE from bestfit950.txt
```

SELECT RECORD assigns a line from the data file to be referenced by the assigned variable name. For example, the following code selects a record from the WideCharMapping section, and the record is accessible by using the MappingData name.

```
SET UnicodeChar to 0x4e00
SELECT RECORD MappingData from WideCharMapping
    where field 1 matches UnicodeChar
```

The following example selects the line.

```
0x4e00 0xa440
```

Values from selected records are referenced by field number. The following example selects the individual data fields from the selected row.

```
SET MultiByteResult to MappingData.Field2
```

In this example, the value of MultiByteResult is the hexadecimal value 0xa440.

```
CODEPAGE 950                ; Chinese (Taiwan, Hong Kong SAR) - ANSI, OEM
CPINFO 2 0x3f 0x003f        ; DBCS CP, Default Char = Question Mark
...
WCTABLE 20321
0x0000 0x0000;    Null
0x0001 0x0001;    Start Of Heading
0x0002 0x0002;    Start Of Text
0x0003 0x0003;    End Of Text
0x0004 0x0004;    End Of Transmission
0x0005 0x0005;    Enquiry
...
0x4e00 0xa440
0x4e01 0xa442
0x4e03 0xa443
0x4e07 0xc94
```

3.1.5.1.1.2 Pseudocode for Mapping a UTF-16 String to a Codepage String

COMMENT This algorithm maps a Unicode string encoded in UTF-16 to a string in the specified ANSI codepage. The supported ANSI codepages are limited to those that can be set as system codepage.

It requires the following externally specified values:

- 1) CodePage: An integer value to represent an ANSI codepage value.

If CodePage value is CP_ACP (0), use the system default ANSI codepage from the OS.

If CodePage value is CP_OEMCP (1), use the system default OEM codepage from the OS.

- 2) UnicodeString: A string encoded in UTF-16. Every Unicode code point is an unsigned 16-bit ("WORD") value. A surrogate pair is not supported in this algorithm.
- 3) UnicodeStringLength: The string length in 16-bit ("WORD") unit for UnicodeString. When UnicodeStringLength is 0, the length is

decided by counting from the beginning of the string to a NULL character (Unicode value U+0000), including the null character.

- 4) MultiByteString: A string encoded in ANSI codepage. Every character can be an 8-bit (byte) unsigned value or two 8-bit unsigned values.
- 5) MultiByteStringLength: The length in bytes, including the byte for NULL terminator. When MultiByteStringLength is 0, the MultiByteString value will not be used in this algorithm. Instead, the length of the result string in ANSI codepage will be returned.
- 6) lpDefaultChar
Optional. Point to the byte to use if a character cannot be represented in the specified codepage. The application sets this parameter to NULL if the function is to use a system default value. The common default value is 0x3f, which is the ASCII value for the question mark.

PROCEDURE WideCharToMultiByteFromCodepageDataFile

```
IF CodePage is CP_ACP THEN
    COMMENT Windows operating system keeps a systemwide value of
        default ANSI system codepage. It is used to provide a default
    COMMENT system codepage to be used by legacy ANSI application.

    SET CodePage to the default ANSI system codepage from the Windows
        operating system.
ELSE IF CodePage is CP_OEMCP THEN
    COMMENT Windows keeps a systemwide value of
        default OEM system codepage. It is used to provide a default
    COMMENT system codepage to be used by legacy console application.

    SET CodePage to the default OEM system codepage from Windows.

ENDIF

IF CodePage is CP_UTF8 THEN
    CALL Utf8ConversionAlgorithm
    COMMENT For UTF-8 use the algorithm in 3.1.5.1.6
    RETURN
ENDIF

IF UnicodeStringLength is 0 THEN
    COMPUTE UnicodeStringLength as the string length in 16-bit units
        of UnicodeString as a NULL-terminated string, including
        NULL terminator.
ENDIF

IF MultiByteStringLength is 0 THEN
    SET IsCountingOnly to True
ELSE
    SET IsCountingOnly to False
ENDIF

SET ResultMultiByteLength to 0
SET CodePageFileName to the concatenation of strings "Bestfit",
    CodePage as a string, and ".txt"

IF lpDefaultChar is null THEN
    COMMENT No default char is specified by the caller. Read the default
    COMMENT char from CPINFO in the data file

    OPEN SECTION CharacterInfo where section name is CPINFO
        from file with the name of CodePageFileName
```

```

        SET lpDefaultChar to CharacterInfo.Field3
    ENDIF

    OPEN SECTION WideCharMapping where section name is WCTABLE from file
        with the name of CodePageFileName

    FOR each Unicode codepoint UnicodeChar in UnicodeString
        SELECT MappingData from WideCharMapping
            where field 1 matches UnicodeChar
        IF MappingData is null THEN
            COMMENT There is no mapping for this Unicode character, use
            COMMENT the default character
            IF IsCountingOnly is False THEN
                SET MultiByteString[ResultMultiByteLength]
                    to lpDefaultChar
            ENDIF
            INCREMENT ResultMultiByteLength
            CONTINUE FOR loop
        ENDIF

        SET MultiByteResult to MappingData.Field2

        IF MultiByteResult is less than 256 THEN
            COMMENT This is a single byte result
            IF IsCountingOnly is True THEN
                INCREMENT ResultMultiByteLength
            ELSE
                SET MultiByteString[ResultMultiByteLength]
                    to MultiByteResult
                INCREMENT ResultMultiByteLength
            ENDIF
        ELSE
            COMMENT This is a double byte result
            IF IsCountingOnly is True THEN
                COMPUTE ResultMultiByteLength as
                    ResultMultiByteLength added by 2
            ELSE
                SET MultiByteString[ResultMultiByteLength] to
                    MultiByteResult divided by 256
                INCREMENT ResultMultiByteLength
                SET MultiByteString[ResultMultiByteLength] to
                    the remainder of MultiByteResult divided by 256
                INCREMENT ResultMultiByteLength
            ENDIF
        ENDIF
    END FOR

    RETURN ResultMultiByteLength as a 32-bit unsigned integer

```

3.1.5.1.1.3 Pseudocode for Mapping a Codepage String to a UTF-16 String

COMMENT This algorithm maps a Unicode string encoded in the specified codepage to UTF-16.

It requires the following externally specified values:

1) CodePage: An integer value to represent an ANSI codepage value.

If CodePage value is CP_ACP (0), use the system default ANSI codepage from the OS.
 If CodePage value is CP_OEMCP (1), use the system default OEM codepage from the OS.
 2) MultiByteString: A string encoded in ANSI codepage. Every character can be an 8-bit (byte) unsigned value or two 8-bit unsigned values.

3) MultiByteStringLength: The length in bytes, including

the byte for terminating null character. When MultiByteStringLength is 0, the length is decided by counting from the beginning of the string to a null character (0x00), including the null character.

- 4) UnicodeString: A string encoded in UTF-16. Every Unicode code point is an unsigned 16-bit ("WORD") value. Surrogate pair is not supported in this algorithm.
- 5) UnicodeStringLength: The string length in 16-bit ("WORD") unit for UnicodeString. When UnicodeStringLength is 0, the UnicodeString value will not be used in this algorithm. Instead, the length of the result string in UTF-16 will be returned.

PROCEDURE MultiByteToWideCharFromCodepageDataFile

```
IF CodePage is CP_ACP THEN
    COMMENT Windows keeps a systemwide value of
        default ANSI system codepage. It is used to provide a default
    COMMENT system codepage to be used by legacy ANSI application.

    SET CodePage to the default ANSI system codepage from Windows.

ELSE IF CodePage is CP_OEMCP THEN
    COMMENT Windows keeps a systemwide value of
        default OEM system codepage. It is used to provide a default
    COMMENT system codepage to be used by legacy console application.

    SET CodePage to the default OEM system codepage from Windows.

ENDIF

IF CodePage is CP_UTF8 THEN
    CALL Utf8ConversionAlgorithm
    COMMENT For UTF-8 use the algorithm in 3.1.5.1.6
    RETURN

ENDIF

IF MultiByteStringLength is 0 THEN
    COMPUTE UnicodeStringLength as the string length in 8-bit units
        of MultiByteString as a null-terminated string, including
        terminating null character.
ENDIF

IF UnicodeStringLength is 0 THEN
    SET IsCountingOnly to True
ELSE
    SET IsCountingOnly to False
ENDIF

SET CodePageFileName to the concatenation of
CodePage as a string, and ".txt"

OPEN SECTION CodePageInfo where section name is CPINFO from file
    with the name of CodePageFileName
COMMENT Read the codepage type.
COMMENT The value for Single Byte Code Page (SBCS) is 1
COMMENT The value for Double Byte Code Page (DBCS) is 2

SET CodePageType to CodePageInfo.Field1
SET DefaultUnicodeChar to CodePageInfo.Field3

OPEN SECTION SingleByteMapping where section name is MBTABLE from file
    with the name of CodePageFileName

SET MultiByteIndex = 0
WHILE MultiByteIndex <= to MultiByteStringLength - 1
```

```

SET MultiByteChar = MultiByteString[MultiByteIndex]
IF CodePageType is 1 THEN
    COMMENT SBCS codepage
    COMMENT Select a record which contains the mapping data
    SELECT MappingData from SingleByteMapping
        where field 1 matches MultiByteChar
    IF MappingData is null THEN
        COMMENT There is no mapping for this single-byte character, use
        COMMENT the default character
        IF IsCountingOnly is False THEN
            SET MultiByteString[ResultUnicodeLength]
                to DefaultUnicodeChar
        ENDIF
        INCREMENT ResultMultiByteLength
        INCREMENT MultiByteIndex
        CONTINUE WHILE loop
    ENDIF
    IF IsCountOnly is False THEN
        SET UnicodeString[ResultUnicodeLength]
            to MappingData.Field2
    ENDIF
    INCREMENT ResultUnicodeLength
ELSE
    COMMENT DBCS codepage
    COMMENT First, try if this is a single-byte mapping
    SELECT MappingData from SingleByteMapping
        where field 1 matches MultiByteChar
    IF MappingData is not null THEN
        COMMENT This byte is a single-byte character
        IF IsCountOnly is False THEN
            SET UnicodeString[ResultUnicodeLength]
                to MappingData.Field2
        ENDIF
        INCREMENT ResultUnicodeLength
    ELSE
        COMMENT Not a single-byte character
        COMMENT Check if this is a valid lead byte for double byte mapping
        OPEN SECTION DBCSRanges
            where section name is DBCSRANGE from file
            with the name of CodePageFileName

        COMMENT Read the count of DBCS Range count
        SET DBCSRangeCount to DBCSRanges.Field1

        SET ValidDBCS to False
        COMMENT Enumerate through every DBCSRange record to see if
        COMMENT the MultiByteChar is a leading byte

        FOR Counter i = 1 to DBCSRangeCount
            COMMENT Select the current record
            SELECT DBCSRangeRecord from DBCSRanges
            SET LeadByteStart to DBCSRangeRecord.Field1
            SET LeadByteEnd to DBCSRangeRecord.Field2
            IF MultiByteChar is larger or equal to LeadByteStart AND
                MultiByteChar is less or equal to LeadByteEnd THEN
                COMMENT This is a valid lead byte
                COMMENT Now check if there is a following valid trailing byte
                SET LeadByteTableCount = MultiByteChar - LeadByteStart

                COMMENT Select the current DBCSTABLE section
                OPEN SECTION DBCSTableSection from DBCSRanges
                    where section name is DBCSTABLE
                COMMENT Advance to the right DBCSTABLE section
                FOR LeadByteIndex = 0 to LeadByteTableCount
                    ADVANCE SECTION DBCSTableSection
                NEXTFOR
                COMMENT Check if the trailing byte is valid
                IF MultiByteIndex + 1 is less than MultiByteStringLength THEN
                    SET TrailByteChar to MultiByteString[MultiByteIndex + 1]

```

```

        SELECT MappingData FROM DBCSTABLE
        Where field 1 matches TrailgByteChar
    IF MappingData is not null THEN
        COMMENT Valid trailing byte
        SET ValidDBCS to True
        IF IsCountingOnly is FALSE THEN
            SET UnicodeString[ResultUnicodeLength] to MappingData.Field2
        ENDIF
        INCREMENT ResultUnicodeLength
        COMMENT Increment the MultiByteIndex.
        COMMENT Note that the MultiByteIndex will
        COMMENT be incremented again for the WHILE loop
        INCREMENT MultiByteIndex
        EXIT FOR loop
    ENDIF
ENDIF
ENDIF
COMMENT No valid lead byte is found. Advance to next record
ADVANCE DBCSRangeRecord
NEXTFOR
IF ValidDBCS is FALSE THEN
    COMMENT There is no valid leading byte/trailing byte sequence
    If IsCountingOnly is FALSE THEN
        SET UnicodeString[ResultUnicodeLength] to DefaultUnicodeChar
    ENDIF
    INCREMENT MultiByteIndex
    INCREMENT ResultUnicodeLength
ENDIF
ENDIF
ENDIF
INCREMENT MultiByteIndex
ENDWHILE

RETURN ResultMultiByteLength as a 32-bit unsigned integer

```

3.1.5.1.2 Mapping Between UTF-16 Strings and ISO 2022-Based Codepages

[\[ECMA-035\]](#) defines the standard that is fully identical with International Standard ISO/IEC 2022:1994. EUC (Extended Unix Code) is based on ISO-2022 standard.

For more information, see [\[ECMA-035\]](#).

3.1.5.1.3 Mapping between UTF-16 Strings and GB 18030 Codepage

Windows implements GB-18030 based on [\[GB18030\]](#).

For more information, please see [\[GB18030\]](#).

3.1.5.1.4 Mapping Between UTF-16 Strings and ISCII Codepage

Windows implements ISCII-based **codepage** based on [\[ISCII\]](#).

For more information, see [\[ISCII\]](#).

3.1.5.1.5 Mapping Between UTF-16 Strings and UTF-7

Windows implements UTF-7 **codepage** based on [\[RFC2152\]](#).

For more information, see [\[RFC2152\]](#).

3.1.5.1.6 Mapping Between UTF-16 Strings and UTF-8

Windows implements UTF-8 **codepage** based on [\[UNICODE5.0.0/CH3\]](#).

For more information, see [UNICODE5.0.0/CH3].

3.1.5.2 Comparing UTF-16 Strings by Using Sort Keys

To compare strings, a **sort key** is required for each string. A binary comparison of the sort keys can then be used to arrange the strings in any order.

3.1.5.2.1 Pseudocode for Comparing UTF-16 Strings

This algorithm compares two **UTF-16** strings by using linguistically appropriate rules.

This algorithm compares two Unicode strings using linguistic appropriate rules. It requires the following externally specified values:

```
1) StringA: A string encoded in UTF-16
2) StringB: A string encoded in UTF-16

CALL GetWindowsSortKey
    WITH StringA
    RETURNING SortKeyA
CALL GetWindowsSortKey
    WITH StringB
    RETURNING SortKeyB

CALL CompareSortKeys
    WITH SortKeyA, SortKeyB
    RETURNING Result

IF Result is "SortKeyA is equal to SortKeyB" THEN
    StringA is considered equal to StringB
ELSE IF Result is "SortKeyA is less than SortKeyB" THEN
    StringA is sorted prior to StringB
ELSE
    StringA is sorted after StringB
ENDIF
```

3.1.5.2.2 CompareSortKey

This algorithm generates **sort keys** for two strings and uses the sort keys to provide a linguistically appropriate string comparison.

```
COMMENT CompareSortKeys
COMMENT On Entry: SortKeyA - An array of bytes returned from
COMMENT                               GetWindowsSortKey
COMMENT                               SortKeyB - An array of bytes returned from
COMMENT                               GetWindowsSortKey
COMMENT
COMMENT On Exit: Result - A value indicating if SortKeyA
COMMENT                  is less than, equal to, or greater
COMMENT                  than SortKeyB

PROCEDURE CompareSortKeys

SET index to 0
WHILE index is less than Length(SortKeyA) and
    index is also less than Length(SortKeyB)

    IF SortKeyA[index] is less than SortKeyB[index] THEN
        SET Result to "SortKeyA is less than SortKeyB"
        RETURN
    ENDIF
```

```

        IF SortKeyA[index] is greater than SortKeyB[index] THEN
            SET Result to "SortKeyA is greater than SortKeyB"
            RETURN
        ENDIF

    INCREMENT index
ENDWHILE

IF Length(SortKeyA) is equal to Length(SortKeyB) THEN
    SET Result to "SortKeyA is equal to SortKeyB"
ELSE IF Length(SortKeyA) is less than Length(SortKeyB) THEN
    SET Result to "SortKeyA is less than SortKeyB"
ELSE
    assert Length(SortKeyA) needs to be greater than Length(SortKeyB)
    SET Result to "SortKeyA is greater than SortKeyB"
ENDIF
RETURN

```

Any sorting mechanism can be used to arrange these strings by comparing their sort keys.

3.1.5.2.3 Accessing the Windows Sorting Weight Table

Windows gets its sorting data from a data table (see section [3.1.5.2.3.1](#)). Code points are labeled by using **UTF-16** values. The file is arranged in sections of tab-delimited field records. Optional comments begin with a semicolon. Each section contains a label and can have a subsection label. [<1>](#)

Note that labels are any field that does not begin with a numerical (0xNNNN) value. Blank lines and characters that follow a ";" are ignored.

This document uses the following notation to specify the processing of the file.

OPEN indicates that queries are made for records in a specific section. To open the preceding section with the SORTKEY label and DEFAULT sublabel, the following syntax is used. The OPEN SECTION is accessible by using the DefaultTable name.

```

OPEN SECTION DefaultTable where name is
    SORTKEY\DEFAULT from unisort.txt

```

SELECT assigns a line from the data file to be referenced by the assigned variable name. To select the highlighted row preceding, this document uses this notation. The selected row is accessible by using the name CharacterRow.

```

SET UnicodeChar to 0x0041
SELECT RECORD CharacterRow FROM DefaultTable
    WHERE field 1 matches UnicodeChar

```

Values from selected records are referenced by field number. The following pseudo code selects the individual data fields from the selected row.

```

SET CharacterWeight.ScriptMember to CharacterRow.Field2
SET CharacterWeight.PrimaryWeight to CharacterRow.Field3
SET CharacterWeight.DiacriticWeight to CharacterRow.Field4
SET CharacterWeight.CaseWeight to CharacterRow.Field5

```

To select the record for characters 0x0043 and 0x0068 with LCID 0x0405, the following notation is used.<2>

```
SET Character1 to 0x0043
SET Character2 to 0x0068
SET SortLocale to 0x0405

OPEN SECTION ContractionTable where name is
    SORTTABLES\COMPRESSION\LCID[SortLocale]\TWO from unisort.txt
SELECT RECORD ContractionRow FROM ContractionTable WHERE field 1
    matches Character1 and field 2 matches Character2
SET CharacterWeight.ScriptMember to ContractionRow.Field3
SET CharacterWeight.PrimaryWeight to ContractionRow.Field4
SET CharacterWeight.DiacriticWeight to ContractionRow.Field5
SET CharacterWeight.CaseWeight to ContractionRow.Field6
```

3.1.5.2.3.1 Windows Sorting Weight Table

This section contains a link to detailed character weight specifications that permit consistent sorting and comparison of Unicode strings. The data is not used by itself but is used as one of the inputs to the comparison algorithm. The layout and format of data in this file is also specified in [\[MSDN-SWT\].<3>](#)

3.1.5.2.4 GetWindowsSortKey Pseudocode

This algorithm specifies the generation of **sort keys** for a specific **UTF-16** string.<4>

```
STRUCTURE CharacterWeightType
(
    ScriptMember:      8 bit integer
    PrimaryWeight:     8 bit integer
    DiacriticWeight:   8 bit integer
    CaseWeight:        8 bit integer
)

STRUCTURE UnicodeWeightType
(
    ScriptMember:      8 bit integer
    PrimaryWeight:     8 bit integer
    ThirdByteWeight:   8 bit integer
)

STRUCTURE SpecialWeightType
(
    Position:          16 bit integer
    ScriptMember:      8 bit integer
    PrimaryWeight:     8 bit integer
)

STRUCTURE ExtraWeightType
(
    W6:                8 bit integer
    W7:                8 bit integer
)

SET constant LCID_KOREAN to 0x0412
SET constant LCID_KOREAN_UNICODE_SORT to 0x010412
SET constant LCID_HUNGARIAN to 0x040e

SET constant SORTKEY_SEPARATOR to 0x01
SET constant SORTKEY_TERMINATOR to 0x00

SET global KoreanScriptMap to InitKoreanScriptMap

//
```

```

// Script Member Values.
//
SET constant UNSORTABLE          to 0
SET constant NONSPACE_MARK      to 1
SET constant EXPANSION          to 2
SET constant EASTASIA_SPECIAL   to 3
SET constant JAMO_SPECIAL       to 4
SET constant EXTENSION_A        to 5
SET constant PUNCTUATION        to 6

SET constant SYMBOL_1           to 7
SET constant SYMBOL_2           to 8
SET constant SYMBOL_3           to 9
SET constant SYMBOL_4           to 10
SET constant SYMBOL_5           to 11
SET constant SYMBOL_6           to 12

SET constant DIGIT               to 13

SET constant LATIN               to 14
SET constant KANA                to 34
SET constant IDEOGRAPH           to 128

IF Windows version is Windows Vista, Windows Server 2008, Windows 7, or
   Windows Server 2008 R2 THEN
SET constant MAX_SPECIAL_CASE to SYMBOL_6

ELSE
SET constant MAX_SPECIAL_CASE to SYMBOL_5
ENDIF
    COMMENT Set the constant for the first script member of the Unicode
    COMMENT Private Use Area (PUA) range
    SET constant PUA3BYTESTART to 0xA9
    COMMENT Set the constant for the last script member of the Unicode
    COMMENT Private Use Area (PUA) range
    SET constant PUA3BYTEEND to 0xAF

    COMMENT Set the constant for the first script member of CJK
    COMMENT (Chinese/Japanese/Korean) 3 byte weight range
    SET constant CJK3BYTESTART to 0xC0
    COMMENT Set the constant for the last script member of CJK
    COMMENT (Chinese/Japanese/Korean) 3 byte weight range
    SET constant CJK3BYTEEND to 0xF9
ENDIF
SET constant FIRST_SCRIPT        to LATIN
SET constant MAX_SCRIPTS         to 256

//
// Values for CJK Unified Ideographs Extension A range.
// 0x3400 thru 0x4dbf
//
SET constant SCRIPT_MEMBER_EXT_A to 254      // SM for Extension A
SET constant PRIMARY_WEIGHT_EXT_A to 255     // AW for Extension A

//
// Lowest weight values.
// Used to remove trailing DW and CW values.
// Also used to keep illegal values out of sort keys.
//

SET constant MIN_DW to 2
SET constant MIN_CW to 2

//
// Bit mask values.
//
// Case Weight (CW) - 8 bits:
// bit 0 => width
// bit 1,2 => small kana, sei-on

```

```

//      bit 3,4 => upper/lower case
//      bit 5   => kana
//      bit 6,7 => contraction
//

SET constant CONTRACTION_8_MASK to 0xc0
SET constant CONTRACTION_7_MASK to 0xc0
SET constant CONTRACTION_6_MASK to 0xc0
SET constant CONTRACTION_5_MASK to 0x80
SET constant CONTRACTION_4_MASK to 0x80
SET constant CONTRACTION_3_MASK to 0x40
SET constant CONTRACTION_2_MASK to 0x40

SET constant CONTRACTION_MASK to 0xc0

ELSE
    COMMENT Otherwise, only 2-character or 3-character contractions
    // are supported.
SET constant CONTRACTION_3_MASK to 0xc0
// Bit-mask to check 2 character contraction or 3 character contraction
SET constant CONTRACTION_2_MASK to 0x80
// Bit-mask to check 2 character contraction
ENDIF

SET constant CASE_UPPER_MASK to 0xe7 // zero out case bits
SET constant CASE_KANA_MASK  to 0xdf // zero out kana bit
SET constant CASE_WIDTH_MASK to 0xfe // zero out width bit

//
// Masks to isolate the various bits in the case weight.
//
// NOTE: Bit 2 needs to always equal 1 to avoid getting
//       a byte value of either 0 or 1.
//

SET constant CASE_EXTRA_WEIGHT_MASK to 0xc4
SET constant ISOLATE_KANA to
    (~CASE_KANA_MASK) | CASE_EXTRA_WEIGHT_MASK
SET constant ISOLATE_WIDTH to
    (~CASE_WIDTH_MASK) | CASE_EXTRA_WEIGHT_MASK

//
// Values for East Asia special case primary weights.
//
SET constant PW_REPEAT      to 0
SET constant PW_CHO_ON      to 1
SET constant MAX_SPECIAL_PW to PW_CHO_ON

//
// Values for weight 5 - East Asia Extra Weights.
//
SET constant WT_FIVE_KANA to 3
SET constant WT_FIVE_REPEAT to 4
SET constant WT_FIVE_CHO_ON to 5

//
// PW Mask for Cho-On:
// Leaves bit 7 on in PW, so it becomes Repeat
// if it follows Kana N.
//
SET constant CHO_ON_PW_MASK to 0x87

//
// Special weight values
//
SET constant MAP_INVALID_WEIGHT to 0xff

//

```

```

// Some Significant Values for Korean Jamo.
// The L, V & T syllables in the 0x1100 Unicode range
// can be composed to characters in the 0xac00 range.
// See The Unicode Standard for details.
//
SET constant NLS_CHAR_FIRST_JAMO          to 0x1100
// Begin Jamo range
SET constant NLS_CHAR_LAST_JAMO           to 0x11f9
// End Jamo range
SET constant NLS_CHAR_FIRST_VOWEL_JAMO    to 0x1160
// First Vowel Jamo
SET constant
    NLS_CHAR_FIRST_TRAILING_JAMO to 0x11a8
// First Trailing Jamo
SET constant
    NLS_JAMO_VOWEL_COUNT to 21
// Number of vowel Jamo (V)
SET constant
    NLS_JAMO_TRAILING_COUNT to 28
// Number of trailing Jamo (L)
SET constant
    NLS_HANGUL_FIRST_COMPOSED to 0xac00
// Begin composed range

//
// Values for Unicode Weight extra weights (e.g. Jamo (old Hangul)).
// The following uses SM for extra UW weights.
//
SET constant ScriptMember Extra UnicodeWeight to 255
// Leading Weight / Vowel Weight / Trailing Weight
// according to the current Jamo class.
//
STRUCTURE JamoSortInfoType
(
    // true for an old Hangul sequence
    OldHangulFlag : Boolean

    // true if U+1160 (Hangul Jungseong Filler) used
    FillerUsed : Boolean

    // index to the prior modern Hangul syllable (L)
    LeadingIndex : 8 bit integer

    // index to the prior modern Hangul syllable (V)
    VowelIndex : 8 bit integer

    // index to the prior modern Hangul syllable (T)
    TrailingIndex : 8 bit integer

    // Weight to offset from other old hangul (L)
    LeadingWeight : 8 bit integer

    // Weight to offset from other old hangul (V)
    VowelWeight : 8 bit integer

    // Weight to offset from other old hangul (T)
    TrailingWeight : 8 bit integer
)

// This is the raw data record type from the data table
STRUCTURE JamoStateDataType
(
    // true for an old Hangul sequence
    OldHangulFlag : Boolean

    // index to the prior modern Hangul syllable (L)
    LeadingIndex : 8 bit integer

    // index to the prior modern Hangul syllable (V)

```

```

VowelIndex : 8 bit integer

// index to the prior modern Hangul syllable (T)
TrailingIndex : 8 bit integer

// weight to distinguish from old Hangul
ExtraWeight : 8 bit integer

// number of additional records in this state
TransitionCount : 8 bit integer

// Current record in unisort.txt Jamo table:
JamoRecord : data record

// SORTTABLES\JAMOSORT\[Character] section
)
COMMENT GetWindowsSortKey
COMMENT
COMMENT On Entry:  SourceString - Unicode String to compute a
COMMENT                               sort key for
COMMENT           SortLocale   - Locale to determine correct
COMMENT                               linguistic sort
COMMENT           Flags        - Bit Flag to control behavior
COMMENT                               of sort key generation.
COMMENT
COMMENT NORM_IGNORENONSPACE      Ignore diacritic weight
COMMENT NORM_IGNORECASE:         Ignore case weight
COMMENT NORM_IGNOREKANATYPE:     Ignore Japanese Katakana/Hiraga
COMMENT                               difference
COMMENT NORM_IGNOREWIDTH:        Ignore Chinese/Japanese/Korean
COMMENT                               half-width and full-width difference.
COMMENT
COMMENT On Exit:  SortKey        - Byte array containing the
COMMENT                               computed sort key.
COMMENT

PROCEDURE GetWindowsSortKey(IN SourceString : Unicode String,
                           IN SortLocale :  LCID,
                           IN Flags : 32 bit integer,
                           OUT SortKey : BYTE String)

COMMENT Compute flags for sort conditions
COMMENT Based on the case/kana/width flags,
COMMENT   turn off bits in case mask when comparing case weight.

SET CaseMask to 0xff

If (NORM IGNORECASE bit is on in Flags) THEN
    SET CaseMask to CaseMask LOGICAL AND with CASE_UPPER_MASK
ENDIF

If (NORM IGNOREKANATYPE bit is on in Flags) THEN
    SET CaseMask to CaseMask LOGICAL AND with CASE_KANA_MASK
ENDIF

If (NORM_IGNOREWIDTH bit is on in Flags) THEN
    SET CaseMask to CaseMask LOGICAL AND with CASE_WIDTH_MASK
ENDIF

COMMENT Windows 7 and Windows Server 2008 R2 use 3-byte
COMMENT (instead of 2-byte) sequence for Unicode Weights
COMMENT for Private Use Area (PUA) and some Chinese/Japanese/Korean (
COMMENT CJK) script members.
COMMENT Does this sort have a 3-byte Unicode Weight (CJK sorts)?
IF Windows version is Windows 7 and Windows Server 2008 R2 THEN
    COMMENT Check if the locale can have 3-byte Unicode weight
    SET Is3ByteWeightLocale to CALL Check3ByteWeightLocale(SortLocale)
ENDIF

```

```

IF Windows version is Windows Vista, Windows Server 2008, Windows 7, or
Windows Server 2008 R2 THEN
    COMMENT For Windows Vista, Windows Server 2008, Windows 7, and
    COMMENT Windows Server 2008 R2, the algorithm
    COMMENT does not remap the script for Korean locale
    SET IsKoreanLocale to false
ELSE

    IF SortLocale is LCID KOREAN or
    SortLocale is LCID_KOREAN_UNICODE_SORT THEN
        SET IsKoreanLocale to true
        IF KoreanScriptMap is null THEN
            CALL InitKoreanScriptMap
        ELSE
            SET IsKoreanLocale to false
        ENDIF
    ENDIF
ENDIF

//
// Allocate buffer to hold different levels of sort key weights.
// UnicodeWeights/ExtraWeights/SpecialWeights will be eventually
// to be collected together, in that order, into the returned
// Sortkey byte string.
//
// Maximum expansion size is 3 times the input size
//

// Unicode Weight => 4 word (16 bit) length
// (extension A and Jamo need extra words)
SET UnicodeWeights to new empty string of UnicodeWeightType

SET DiacriticWeights to new empty string of BYTE
SET CaseWeights to new empty string of BYTE

// Extra Weight=>4 byte length (4 weights, 1 byte each) FE Special
SET ExtraWeights to new empty string of ExtraWeightType

// Special Weight => dword length (2 words each of 16 bits)
SET SpecialWeights to new empty string of SpecialWeightType

//
// Go through the string, code point by code point,
// testing for contractions and Hungarian special character sequence
//

// loop presumes 0 based index for source string
FOR SourceIndex is 0 to Length(SourceString) -1
    //
    // Get weights
    // CharacterWeight will contain all of the weight information
    // for the character tested.
    //

    SET CharacterWeight to CALL GetCharacterWeights
    WITH (SortLocale, SourceString[SourceIndex])

    SET ScriptMember to CharacterWeight.ScriptMember

    // Special case weights have script members less than
    // MAX_SPECIAL_CASE (11)
    IF ScriptMember is greater than MAX_SPECIAL_CASE THEN

        //
        // No special case on character, but has to check for
        // contraction characters and Hungarian special
        // character sequence characters.
        //

```

```

SET HasHungarianSpecialCharacterSequence to CALL
  TestHungarianCharacterSequences
  WITH (SortLocale, SourceString, SourceIndex)

SET Result to CALL GetContractionType WITH (CharacterWeight)

CASE Result OF

  "3-character Contraction":
    COMMENT This is only possible for Windows versions that
    COMMENT are Windows NT 4.0 through Windows Server 2003
    Set ContractionFound to CALL SortkeyContractionHandler
      WITH (SortLocale, SourceString, SourceIndex,
        HasHungarianSpecialCharacterSequence, 3,
        UnicodeWeights, DiacriticWeights, CaseWeights)
    IF ContractionFound is true THEN
      COMMENT Break out of the case statement
      BREAK
    ENDIF
    IF ContractionFound is true THEN
      COMMENT Break out of the case statement
      BREAK
    ENDIF
    COMMENT If no contraction is found, fall through into additional cases.
    FALLTHROUGH

  "2-character Contraction":
    COMMENT This is only possible for Windows versions that are
    COMMENT Windows NT 4.0 through Windows Server 2003
    Set ContractionFound to CALL SortkeyContractionHandler
      WITH (SortLocale, SourceString, SourceIndex,
        HasHungarianSpecialCharacterSequence, 2,
        UnicodeWeights, DiacriticWeights, CaseWeights)
    IF ContractionFound is true THEN
      COMMENT Break out of the case statement
      BREAK
    ENDIF
    COMMENT If no contraction is found, fall through into the OTHER case.
    COMMENT Since "3-character contraction" or "2-character contraction"
    COMMENT are the only two possible values for
    COMMENT Windows NT 4.0 through Windows Server 2003, all calls to
    COMMENT SortkeyContractionHandler will return false.
    COMMENT So, the fallthrough will go directly to the OTHERS section
    FALLTHROUGH

  "6-character contraction, 7-character contraction, or 8-character contraction":
    Set ContractionFound to CALL SortkeyContractionHandler
      WITH (SortLocale, SourceString, SourceIndex,
        HasHungarianSpecialCharacterSequence, 8,
        UnicodeWeights, DiacriticWeights, CaseWeights)
    IF ContractionFound is true THEN
      COMMENT Break out of the case statement
      BREAK
    ELSE
      Set ContractionFound to CALL SortkeyContractionHandler
      WITH (SortLocale, SourceString, SourceIndex,
        HasHungarianSpecialCharacterSequence, 7,
        UnicodeWeights, DiacriticWeights, CaseWeights)
    ENDIF
    IF ContractionFound is true THEN
      COMMENT Break out of the case statement
      BREAK
    ELSE
      Set ContractionFound to CALL SortkeyContractionHandler
      WITH (SortLocale, SourceString, SourceIndex,
        HasHungarianSpecialCharacterSequence, 6,
        UnicodeWeights, DiacriticWeights, CaseWeights)
    ENDIF

```

```

    IF ContractionFound is true THEN
        COMMENT Break out of the case statement
        BREAK
    ENDIF
    COMMENT If no contraction is found, fall through into additional cases.
    FALLTHROUGH

"4-character contraction or 5-character contraction":
    Set ContractionFound to CALL SortkeyContractionHandler
    WITH (SortLocale, SourceString, SourceIndex,
        HasHungarianSpecialCharacterSequence, 5,
        UnicodeWeights, DiacriticWeights, CaseWeights)
    IF ContractionFound is true THEN
        COMMENT Break out of the case statement
        BREAK
    ELSE
        Set ContractionFound to CALL SortkeyContractionHandler
        WITH (SortLocale, SourceString, SourceIndex,
            HasHungarianSpecialCharacterSequence, 4,
            UnicodeWeights, DiacriticWeights, CaseWeights)
    ENDIF
    IF ContractionFound is true THEN
        COMMENT Break out of the case statement
        BREAK
    ENDIF
    COMMENT If no contraction is found, fall through into additional cases.
    FALLTHROUGH

"2-character contraction or 3-character contraction":
    Set ContractionFound to CALL SortkeyContractionHandler
    WITH (SortLocale, SourceString, SourceIndex,
        HasHungarianSpecialCharacterSequence, 3,
        UnicodeWeights, DiacriticWeights, CaseWeights)
    IF ContractionFound is true THEN
        COMMENT Break out of the case statement
        BREAK
    ELSE
        Set ContractionFound to CALL SortkeyContractionHandler
        WITH (SortLocale, SourceString, SourceIndex,
            HasHungarianSpecialCharacterSequence, 2,
            UnicodeWeights, DiacriticWeights, CaseWeights)
    ENDIF
    IF ContractionFound is true THEN
        COMMENT Break out of the case statement
        BREAK
    ENDIF
    COMMENT If no contraction is found, fall through into additional cases.
    FALLTHROUGH

OTHERS :
    IF Windows version is greater than Windows Server 2008 R2 or Windows 7
    THEN
        COMMENT In Windows Server 2008 R2 or Windows 7,
        COMMENT Private Use Area (PUA) code points
        COMMENT and some CJK (Chinese/Japanese/Korean) sorts
        COMMENT might need 3 byte weights
        COMMENT Store normal Unicode weight first. Note that there is no
        COMMENT adjustment of Korean weight anymore.
        SET UnicodeWeight to
            CorrectUnicodeWeight(CharacterWeight, FALSE)
        COMMENT Assume 3-byte Unicode Weight is not used first.
        COMMENT The algorithm will check this later.
        SET UnicodeWeight.ThirdByteWeight to 0

        IF (ScriptMember is equal to or greater than PUA3BYTESTART)
        AND
        (ScriptMember is less than or equal to PUA3BYTEEND) THEN
            SET IsScriptMemberPUA3BYTEWeight to true

```

```

ELSE
    SET IsScriptMemberPUA3ByteWeight to false
ENDIF

IF (ScriptMember is equal to or greater than CJK3BYTESTART) AND
   (ScriptMember is less than or equal to CJK3BYTEEND) THEN
    SET IsScriptMemberCJK3ByteWeight to true
ELSE
    SET IsScriptMemberCJK3ByteWeight to false
ENDIF
IF (IsScriptMemberPUA3ByteWeight is true) OR
   (Is3ByteWeightLocale AND
    IsScriptMemberCJK3ByteWeight is true) THEN
    COMMENT PUA code points and some CJK sorts need 3 byte weights
    SET UnicodeWeight.ThirdByteWeight to CharacterWeight.DiacriticWeight
ELSE
    COMMENT Normal Diacritic Weight
    APPEND CharacterWeight.DiacriticWeight to DiacriticWeights as a BYTE
ENDIF
APPEND UnicodeWeight to UnicodeWeights

SET CaseWeight to GetCaseWeight(CharacterWeight)
APPEND CharacterWeight.CaseWeight to CaseWeights as a BYTE

ELSE

    SET UnicodeWeight to
        CorrectUnicodeWeight(CharacterWeight, IsKoreanLocale)
    APPEND UnicodeWeight to UnicodeWeights
    APPEND CharacterWeight.DiacriticWeight to DiacriticWeights
        as a BYTE
    SET CaseWeight to GetCaseWeight(CharacterWeight)
    APPEND CharacterWeight.CaseWeight to CaseWeights as a BYTE
ENDIF
ENDCASE
ELSE
    CALL SpecialCaseHandler WITH (SourceString, SourceIndex,
        UnicodeWeights, ExtraWeights, SpecialWeights,
        SortLocale, IsKoreanLocale)
ENDIF
ENDFOR

//
// Store the Unicode Weights in the destination buffer.
//
FOR each UnicodeWeight in UnicodeWeights
    //
    // Copy Unicode weight to destination buffer.
    //
    APPEND UnicodeWeight.ScriptMember to SortKey as a BYTE
    APPEND UnicodeWeight.PrimaryWeight to SortKey as a BYTE
    IF Windows version is greater than Windows Server 2008 R2 or Windows 7 THEN
        IF UnicodeWeight.ThirdByteWeight is not 0 THEN
            COMMENT When 3-byte Unicode Weight is used, append the additional
            COMMENT BYTE into SortKey
            APPEND UnicodeWeight.ThirdByteWeight to SortKey as a BYTE
        ENDIF
    ENDIF
ENDFOR

//
// Copy Separator to destination buffer.
//
APPEND SORTKEY_SEPARATOR to SortKey as a BYTE

//

```

```

// Store Diacritic Weights in the destination buffer.
//
IF (NORM_IGNORENONSPACE bit is not turned on in Flags) THEN
    IF (IsReverseDW is TRUE) THEN
        //
        // Reverse diacritics:
        // - remove diacritics from left to right.
        // - store diacritics from right to left.
        //
        FOR each DiacriticWeight in
            DiacriticWeights in the "first in first out" order
            IF DiacriticWeight <= MIN_DW THEN
                REMOVE DiacriticWeight from DiacriticWeights
            ELSE
                BREAK from the current FOR loop
            ENDIF
        ENDFOR

        FOR each DiacriticWeight in
            DiacriticWeights in the "last in first out" order
            //
            // Copy Unicode weight to destination buffer.
            //
            APPEND DiacriticWeight to SortKey as a BYTE
        ENDFOR
    ELSE
        //
        // Regular diacritics:
        // - remove diacritics from right to left.
        // - store diacritics from left to right.
        //
        FOR each DiacriticWeight in
            DiacriticWeights in the "last in first out" order
            IF DiacriticWeight <= MIN_DW THEN
                REMOVE DiacriticWeight from DiacriticWeights
            ELSE
                BREAK from the current FOR loop
            ENDIF
        ENDFOR

        FOR each DiacriticWeight in
            DiacriticWeights in the order of "first in first out"
            //
            // Copy Unicode weight to destination buffer.
            //
            APPEND DiacriticWeight to SortKey as a BYTE
        ENDFOR
    ENDIF
ENDIF

//
// Copy Separator to destination buffer.
//
APPEND SORTKEY_SEPARATOR to SortKey as a BYTE

//
// Store case Weights
//
// - Eliminate minimum CW.
// - Copy case weights to destination buffer.
//
IF (NORM_IGNORECASE bit is not turned on in Flags
    OR NORM_IGNOREWIDTH bit is not turned on in Flags) THEN
    FOR each CaseWeight in CaseWeights
        in the "last in first out" order
        IF CaseWeight <= MIN_CW THEN
            REMOVE CaseWeight from CaseWeights
        ELSE
            BREAK from the current FOR loop
        ENDIF
    ENDIF

```

```

        ENDFOR

        FOR each CaseWeight in CaseWeights
            //
            // Copy Unicode weight to destination buffer.
            //
            APPEND CaseWeight to SortKey as a BYTE
        ENDFOR
    ENDIF

    //
    // Copy Separator to destination buffer.
    //
    APPEND SORTKEY_SEPARATOR to SortKey as a BYTE

    //
    // Store the Extra Weights in the destination buffer for
    // EAST ASIA Special.
    //
    // - Eliminate unnecessary XW.
    // - Copy extra weights to destination buffer.
    //
    IF Length(ExtraWeights) is greater than 0 THEN
        IF (NORM_IGNORENONSPACE bit is turned on in Flag) THEN
            APPEND 0xff to SortKey as a BYTE
            APPEND 0x02 to SortKey as a BYTE
        ENDIF

        // Append W6 group to SortKey
        // Trim unused values from the end of the string
        SET EndExtraWeight to Length(ExtraWeights) - 1

        WHILE EndExtraWeight greater than 0 and
            ExtraWeightSeparator[EndExtraWeight].W6 == 0xe4
            DECREMENT EndExtraWeight
        ENDWHILE

        SET ExtraWeightIndex to 0
        WHILE ExtraWeightIndex is less than or equal to EndExtraWeight
            APPEND ExtraWeightSeparator[ExtraWeightIndex].W6
            to SortKey as a BYTE
            INCREMENT ExtraWeightIndex
        ENDWHILE

        // Append W6 separator
        APPEND 0xff to SortKey as a BYTE

        // Append W7 group to SortKey
        // Trim unused values from the end of the string
        SET EndExtraWeight to Length(ExtraWeights) - 1
        WHILE EndExtraWeight greater than 0 and
            ExtraWeightSeparator[EndExtraWeight].W7 == 0xe4
            DECREMENT EndExtraWeight
        ENDWHILE

        SET ExtraWeightIndex to 0
        WHILE ExtraWeightIndex is less than or equal to EndExtraWeight
            APPEND ExtraWeightSeparator[ExtraWeightIndex].W7 to SortKey
            INCREMENT ExtraWeightIndex
        ENDWHILE

        // Append W7 separator
        APPEND 0xff to SortKey as a BYTE
    ENDIF

    //
    // Copy Separator to destination buffer.
    //
    APPEND SORTKEY_SEPARATOR to SortKey as a BYTE

```

```

//
// Store the Special Weights in the destination buffer.
//
// - Copy special weights to destination buffer.
//
FOR each SpecialWeight in SpecialWeights
    // High byte (most significant)
    SET Byte1 to SpecialWeight.Position >> 8
    // Low byte (least significant)
    SET Byte2 to SpecialWeight.Position & 0xff
    APPEND Byte1 to SortKey as a BYTE
    APPEND Byte2 to SortKey as a BYTE
    APPEND SpecialWeight.Script to SortKey as a BYTE
    APPEND SpecialWeight.Weight to SortKey as a BYTE
ENDFOR

//
// Copy terminator to destination buffer.
//
APPEND SORTKEY_TERMINATOR to SortKey

RETURN SortKey

```

3.1.5.2.5 TestHungarianCharacterSequences

This algorithm checks if the specified **UTF-16** string has a Hungarian special-character sequence for the specified locale in the specific string index.

Hungarian contains special character sequences in which the first character of the string designates a string that is equivalent to the last three characters of the string. For example, the string "ddzs" is actually treated as the string "dzsdzs" for the purposes of generating the sort key. This function checks to see if the specified locale is Hungarian, and it also checks to see if the next two characters starting in the specified index are the same. If so, this indicates that it is a likely Hungarian special-character sequence.

```

COMMENT TestHungarianCharacterSequences
COMMENT
COMMENT On Entry:  SortLocale  - Locale to use for linguistic data
COMMENT              SourceString - Unicode String to look for Hungarian
COMMENT                      special character sequence in
COMMENT              SourceIndex - Index of character in string to
COMMENT                      look for start of
COMMENT                      Hungarian special character sequence
COMMENT
COMMENT On Exit:   Result      - Set to true if a Hungarian special
COMMENT                      character sequence
COMMENT                      was found
COMMENT

PROCEDURE TestHungarianCharacterSequences(IN SortLocale : LCID,
                                           IN SourceString : Unicode String,
                                           IN SourceIndex : 32 bit integer,
                                           OUT Result : Boolean)

    // Hungarian special character sequence only happen to Hungarian
    // Note that this can be found in unisort.txt in the
    // SORTTABLES\DOUBLECOMPRESSION section, however since
    // there's only 1 locale just hard code it here.
    IF SortLocale not equal to LCID HUNGARIAN) THEN
        SET Result to false
        RETURN
    ENDIF

    // first test to make sure more data is available

```

```

IF SourceIndex + 1 is greater than or equal to
    Length(SourceString) THEN
    SET Result to false
    RETURN
ENDIF

// CMP_MASKOFF_CW (e7) is not necessary
// since it was already masked off
SET FirstWeight to CALL GetCharacterWeights WITH
    (SortLocale, SourceString[SourceIndex])
SET SecondWeight to CALL GetCharacterWeights WITH
    (SortLocale, SourceString[SourceIndex + 1])

IF FirstWeight is equal to SecondWeight THEN
    SET Result to true
ELSE
    SET Result to false
ENDIF

RETURN

```

3.1.5.2.6 GetContractionType

This algorithm specifies the checking of the type of contraction based on the character weight. Contraction is defined by [\[UNICODE-COLLATION\]](#) section 3.2.

For instance, "ll" acts as a single unit in Spanish so that it comes between l and m. This is a two-character contraction. Similarly, "dzs" acts as a single unit in Hungarian, so it is a three-character contraction.

These functions specify if the weights will not be at the beginning of a contraction, the beginning of a two-character contraction, or the beginning of a three-character contraction.

```

COMMENT GetContractionType
COMMENT
COMMENT On Entry:  CharacterWeight - Weights structure to test for
COMMENT                                     a contraction
COMMENT
COMMENT On Exit:   Result           - Type of contraction found:
COMMENT                                     "No contraction"
COMMENT                                     "3-character contraction"
COMMENT                                     "2-character contraction"
COMMENT                                     The following results are only possible for
COMMENT                                     Windows Vista, Windows Server 2008, Windows 7, and
COMMENT                                     Windows Server 2008 R2
COMMENT                                     "6-character contraction, 7-character contraction or
COMMENT                                     8-character contraction"
COMMENT                                     "4-character contraction or 5-character contraction"
COMMENT                                     "2-character contraction or 3-character contraction"

PROCEDURE GetContractionType(IN CharacterWeight : CharacterWeightType,
    OUT Result)
    IF Windows version is Windows NT 4.0 to Windows 2003 THEN
        CASE CharacterWeight.CaseWeight & CONTRACTION_3_MASK OF
            CONTRACTION_3_MASK : SET Result = "3-character contraction"
            CONTRACTION_2_MASK : SET Result = "2-character contraction"
            OTHERS : SET Result = "No contraction"
        ENDCASE
    ELSE
        COMMENT Windows Vista, Windows Server 2008, Windows 7, and Windows Server 2008 R2
        CASE CharacterWeight.CaseWeight & CONTRACTION_MASK OF
            CONTRACTION_6_MASK : SET Result = "6-character contraction, 7-
            character contraction or 8-character contraction"
            CONTRACTION_4_MASK : SET Result = "4-character contraction or 5-

```

```

        character contraction"
        CONTRACTION_2_MASK : SET Result = "2-character contraction or 3-
        character contraction"
        OTHERS : SET Result = "No contraction"
    ENDCASE
ENDIF

RETURN

```

3.1.5.2.7 CorrectUnicodeWeight

This algorithm specifies the processing of the corrected Unicode weight for the specific character weight, and whether the locale is a Korean locale.

```

COMMENT CorrectUnicodeWeight
COMMENT
COMMENT On Entry:  CharacterWeight - Weights structure to get Unicode
COMMENT                                     weight of
COMMENT                IsKoreanLocale - True if this locale needs
COMMENT                                     adjustment for
COMMENT                                     Korean mapped scripts behavior.
COMMENT
COMMENT On Exit:   UnicodeWeight - Corrected Unicode Weight
COMMENT

PROCEDURE
    CorrectUnicodeWeight(IN CharacterWeight : CharacterWeightType,
                        IN IsKoreanLocale : boolean,
                        OUT UnicodeWeight : UnicodeWeightType)

SET UnicodeWeight to CALL MakeUnicodeWeight WITH
    (CharacterWeight.ScriptMember, CharacterWeight.PrimaryWeight,
     IsKoreanLocale)

RETURN UnicodeWeight

```

3.1.5.2.8 MakeUnicodeWeight

This algorithm specifies the generation of the Unicode weight based on the script member, the primary weight, and whether the locale is a Korean locale.

```

COMMENT MakeUnicodeWeight
COMMENT
COMMENT On Entry:  ScriptMember - Script member to use for
COMMENT                                     Unicode weight
COMMENT                PrimaryWeight - Primary weight to use for
COMMENT                                     Unicode weight
COMMENT                IsKoreanLocale - True if this locale needs
COMMENT                                     adjustment for Korean mapped
COMMENT                                     scripts behavior.
COMMENT
COMMENT On Exit:   UnicodeWeight - Corrected Unicode Weight
COMMENT

PROCEDURE MakeUnicodeWeight(IN ScriptMember : 8 bit byte,
                          IN PrimaryWeight : 8 bit byte,
                          IN IsKoreanLocale : boolean,
                          OUT UnicodeWeight : UnicodeWeightType)

IF IsKoreanLocale is true THEN
    SET UnicodeWeight.ScriptMember to
        KoreanScriptMap[ScriptMember]
ELSE
    SET UnicodeWeight.ScriptMember to ScriptMember

```

```

ENDIF

SET UnicodeWeight.PrimaryWeight to PrimaryWeight
RETURN UnicodeWeight

```

3.1.5.2.9 GetCharacterWeights

This algorithm specifies the retrieval of the character weight based on the specified locale and the specified **UTF-16** code point.

```

COMMENT GetCharacterWeights
COMMENT
COMMENT On Entry:  SortLocale      - Locale to use for linguistic
COMMENT                               data
COMMENT                SourceCharacter - Unicode Character to return
COMMENT                               weight for
COMMENT
COMMENT On Exit:   Result          - A structure containing the
COMMENT                               weights for this character
COMMENT

PROCEDURE GetCharacterWeights(IN SortLocale : LCID,
                             IN SourceCharacter : Unicode Character,
                             OUT Result : CharacterWeightType)

// Search for the character in the exception table
OPEN SECTION ExceptionTable where name is
    SORTTABLES\EXCEPTION\LCID[SortLocale] from unisort.txt
SELECT RECORD CharacterRow FROM ExceptionTable WHERE field 1
    matches SourceCharacter

IF CharacterRow is null THEN
    // Not found, search for the character in the default table
    OPEN SECTION DefaultTable where name is
        SORTKEY\DEFAULT from unisort.txt
    SELECT RECORD CharacterRow FROM DefaultTable where field 1
        matches SourceCharacter

    IF CharacterRow is null THEN
        // Not found in default table either, check expansions
        SET Expansion to GetExpandedCharacters(SourceCharacter)

        IF Expansion is not null THEN
            // Has an expansion, set appropriate weights
            SET Result.ScriptMember to EXPANSION
        ELSE
            // No expansion, set appropriate weights
            SET Result.ScriptMember to UNSORTABLE
        ENDIF

        SET Result.PrimaryWeight to 0
        SET Result.DiacriticWeight to 0
        SET Result.CaseWeight to 0

        RETURN Result
    ENDIF
ENDIF

SET Result.ScriptMember to CharacterRow.Field2
SET Result.PrimaryWeight to CharacterRow.Field3
SET Result.DiacriticWeight to CharacterRow.Field4
SET Result.CaseWeight to CharacterRow.Field5

RETURN Result

```

3.1.5.2.10 GetExpansionWeights

This algorithm specifies the generation of a character weight for the specified character that has the expansion behavior, as defined in [\[UNICODE-COLLATION\]](#) section 3.2.

```
COMMENT GetExpansionWeights
COMMENT
COMMENT On Entry: SourceCharacter - Character to look up
COMMENT                               expansions for
COMMENT SortLocale - Locale to get sort weights for
COMMENT
COMMENT On Exit:  Weights - String of 2 or 3 weights for
COMMENT                               this character
COMMENT

PROCEDURE GetExpansionWeights(IN SourceCharacter : Unicode Character,
                              IN SortLocale : LCID,
                              OUT Weights : CharacterWeightType String)

SET Weights to new empty string of CharacterWeightType
SET ExpandedCharacters to CALL GetExpandedCharacters WITH
    (SourceCharacter)

// Append first weight
SET Weight to CALL GetCharacterWeights WITH
    (SortLocale, ExpandedCharacters[0])
APPEND Weight to Weights

// Get second weight, it might expand again
SET Weight to CALL GetCharacterWeights WITH
    (SortLocale, ExpandedCharacters[1])

IF Weight.ScriptMember is EXPANSION THEN
    // second weight expands again, get new expansion
    // note that this can only happen once, as it does
    // with the U=fb03 (ffi ligature)

    SET ExpandedCharacters to CALL
        GetExpandedCharacters(ExpandedCharacters[1])

    // Append second expansion's first weight
    SET Weight to CALL GetCharacterWeights WITH
        (SortLocale, ExpandedCharacters[0])
    APPEND Weight to Weights

    // Get second weight for second expansion, it will not expand again
    SET Weight to CALL GetCharacterWeights WITH
        (SortLocale, ExpandedCharacters[1])
ENDIF

// Finish appending second weight to weights string
APPEND Weight to Weights

RETURN Result
```

3.1.5.2.11 GetExpandedCharacters

This algorithm specifies the generation of the array of expanded characters, if the specified character can be expanded.

```
COMMENT GetExpandedCharacters
COMMENT
COMMENT On Entry: SourceCharacter - Character to look for in
COMMENT                               expansion table
COMMENT
COMMENT On Exit:  Result - Array of two unicode characters
```

```

COMMENT                                for the expansion or null if no
COMMENT                                expansion found
COMMENT
COMMENT NOTE: Look for default table characters first, some entries
COMMENT      in the expansion table are only used in exception tables
COMMENT      for some locales (ie: 0x00c4 Å)

PROCEDURE
    GetExpandedCharacters(IN SourceCharacter : Unicode Character,
                          OUT Result : Unicode Character[2])

// Search for the expansion in the expansion table
OPEN SECTION ExpansionTable where name is
    SORTTABLES\EXPANSION from unisort.txt

SELECT RECORD ExpansionRow FROM ExceptionTable WHERE field 1
    matches SourceCharacter

IF ExpansionRow is null THEN
    SET Result to null
    RETURN Result
ENDIF

SET Result[0] to ExpansionRow.Field2
SET Result[1] to ExpansionRow.Field3

RETURN Result

```

3.1.5.2.12 SortkeyContractionHandler

This algorithm checks if the next few characters in the specified string and index have an 8-character, 7-character, 6-character, 5-character, 4-character, 3-character, or 2-character contraction sequence. If true, these characters are given just one character weight. This algorithm also handles the Hangiran special character sequence.

```

COMMENT SortkeyContractionHandler
COMMENT
COMMENT On Entry: SourceString - Source Unicode String
COMMENT              SourceIndex - Current index within source string
COMMENT              HasHungarianSpecialCharacterSequence: Is the character that the current
COMMENT                  index points to
COMMENT                  the starting of the Hungarian special character sequence
COMMENT              ContractionType: The contraction type, from 2-character to 8-character
COMMENT                  contraction, to be checked against
COMMENT              UnicodeWeights - String of UnicodeWeightType to
COMMENT                  append additional weight(s) to
COMMENT              DiacriticWeights - String of Diacritic Weight to
COMMENT                  append extra weight(s) to if
COMMENT                  needed
COMMENT              CaseWeights - String of Case Weight to
COMMENT                  append special weight(s) to
COMMENT                  if needed
COMMENT
COMMENT On Exit: Result: a string to indicate the type of contraction from the specified
COMMENT              string
COMMENT              UnicodeWeights - The UnicodeWeight of the
COMMENT                  processed character(s) is
COMMENT                  appended to this string.
COMMENT              DiacriticWeights - The Diacritic weight, if any, of
COMMENT                  the processed character(s) is
COMMENT                  appended to this string.
COMMENT              CaseWeights - The Case Weight, if any,
COMMENT                  of the processed character(s)
COMMENT                  is appended to this string.
COMMENT
PROCEDURE SortkeyContractionHandler (IN SortLocale: LCID,

```

```

    IN SourceString: Unicode String,
    IN SourceIndex: 32-bit integer,
    IN HasHungarianSpecialCharacterSequence: boolean
    IN ContractionType: integer number from 2 to 8
    INOUT UnicodeWeights: string of UnicodeWeightType
    INOUT DiacriticWeights: string of BYTE
    INOUT CaseWeights: string of BYTE)

Result: CharacterWeightType

IF HasHungarianSpecialCharacterSequence is true THEN
    COMMENT The beginning of Hungarian special character sequence,
    COMMENT advance one character before starting to check for contraction sequence
    SET SourceIndex to SourceIndex + 1
ENDIF

IF SourceIndex + ContractionType is greater than or equal to SourceString.Length THEN
    SET Result to null
    RETURN false
ENDIF

COMMENT Search for the character in the character contraction table
COMMENT Search for contraction section based on ContractionType

CASE ContractionType
    "8":
    OPEN SECTION ContractionTable where name is
        SORTTABLES\COMPRESSION\LCID[SortLocale]\EIGHT from unisort.txt
    "7":
    OPEN SECTION ContractionTable where name is
        SORTTABLES\COMPRESSION\LCID[SortLocale]\SEVEN from unisort.txt
    "6":
    OPEN SECTION ContractionTable where name is
        SORTTABLES\COMPRESSION\LCID[SortLocale]\SIX from unisort.txt
    "5":
    OPEN SECTION ContractionTable where name is
        SORTTABLES\COMPRESSION\LCID[SortLocale]\FIVE from unisort.txt
    "4":
    OPEN SECTION ContractionTable where name is
        SORTTABLES\COMPRESSION\LCID[SortLocale]\FOUR from unisort.txt
    "3":
    OPEN SECTION ContractionTable where name is
        SORTTABLES\COMPRESSION\LCID[SortLocale]\THREE from unisort.txt
    "2":
    OPEN SECTION ContractionTable where name is
        SORTTABLES\COMPRESSION\LCID[SortLocale]\TWO from unisort.txt

ENDCASE

COMMENT Contraction table might not be found if locale doesn't have them
IF ContractionTable is null THEN
    SET Result to null
    RETURN false
ENDIF

CASE ContractionType
    "8":
        SELECT RECORD ContractionRow FROM ContractionTable
            WHERE field 1 matches SourceString[SourceIndex] and
            WHERE field 2 matches SourceString[SourceIndex + 1] and
            WHERE field 3 matches SourceString[SourceIndex + 2] and
            WHERE field 4 matches SourceString[SourceIndex + 3] and
            WHERE field 5 matches SourceString[SourceIndex + 4] and
            WHERE field 6 matches SourceString[SourceIndex + 5] and
            WHERE field 7 matches SourceString[SourceIndex + 6] and
            WHERE field 8 matches SourceString[SourceIndex + 7]

        COMMENT If this sequence isn't a contraction then one will not be found

```

```

    IF ContractionRow is null THEN
        SET Result to null
    RETURN false
    ENDIF

    COMMENT Found a contraction, get its weights
    SET Result.ScriptMember to ContractionRow.Field9
    SET Result.PrimaryWeight to ContractionRow.Field10

    SET Result.DiacriticWeight to ContractionRow.Field11
    SET Result.CaseWeight to ContractionRow.Field12
"7":
    SELECT RECORD ContractionRow FROM ContractionTable
        WHERE field 1 matches SourceString[SourceIndex] and
        WHERE field 2 matches SourceString[SourceIndex + 1] and
        WHERE field 3 matches SourceString[SourceIndex + 2] and
        WHERE field 4 matches SourceString[SourceIndex + 3] and
        WHERE field 5 matches SourceString[SourceIndex + 4] and
        WHERE field 6 matches SourceString[SourceIndex + 5] and
        WHERE field 7 matches SourceString[SourceIndex + 6]

    COMMENT If this sequence isn't a contraction then one will not be found
    IF ContractionRow is null THEN
        SET Result to null
        RETURN false
    ENDIF

    COMMENT Found a contraction, get its weights
    SET Result.ScriptMember to ContractionRow.Field8
    SET Result.PrimaryWeight to ContractionRow.Field9

    SET Result.DiacriticWeight to ContractionRow.Field10
    SET Result.CaseWeight to ContractionRow.Field11

"6":
    SELECT RECORD ContractionRow FROM ContractionTable
        WHERE field 1 matches SourceString[SourceIndex] and
        WHERE field 2 matches SourceString[SourceIndex + 1] and
        WHERE field 3 matches SourceString[SourceIndex + 2] and
        WHERE field 4 matches SourceString[SourceIndex + 3] and
        WHERE field 5 matches SourceString[SourceIndex + 4] and
        WHERE field 6 matches SourceString[SourceIndex + 5]

    COMMENT If this sequence isn't a contraction then one will not be found
    IF ContractionRow is null THEN
        SET Result to null
        RETURN false
    ENDIF

    COMMENT Found a contraction, get its weights
    SET Result.ScriptMember to ContractionRow.Field7
    SET Result.PrimaryWeight to ContractionRow.Field8

    SET Result.DiacriticWeight to ContractionRow.Field9
    SET Result.CaseWeight to ContractionRow.Field10

"5":
    SELECT RECORD ContractionRow FROM ContractionTable
        WHERE field 1 matches SourceString[SourceIndex] and
        WHERE field 2 matches SourceString[SourceIndex + 1] and
        WHERE field 3 matches SourceString[SourceIndex + 2] and
        WHERE field 4 matches SourceString[SourceIndex + 3] and
        WHERE field 5 matches SourceString[SourceIndex + 4]

    COMMENT If this sequence isn't a contraction then one will not be found
    IF ContractionRow is null THEN
        SET Result to null
        RETURN false
    ENDIF

```

```

COMMENT Found a contraction, get its weights
SET Result.ScriptMember to ContractionRow.Field6
SET Result.PrimaryWeight to ContractionRow.Field7

SET Result.DiacriticWeight to ContractionRow.Field8
SET Result.CaseWeight to ContractionRow.Field9

"4":
SELECT RECORD ContractionRow FROM ContractionTable
    WHERE field 1 matches SourceString[SourceIndex] and
    WHERE field 2 matches SourceString[SourceIndex + 1] and
    WHERE field 3 matches SourceString[SourceIndex + 2] and
    WHERE field 4 matches SourceString[SourceIndex + 3]

COMMENT If this sequence isn't a contraction then one will not be found
IF ContractionRow is null THEN
    SET Result to null
    RETURN false
ENDIF

COMMENT Found a contraction, get its weights
SET Result.ScriptMember to ContractionRow.Field5
SET Result.PrimaryWeight to ContractionRow.Field6

SET Result.DiacriticWeight to ContractionRow.Field7
SET Result.CaseWeight to ContractionRow.Field8

"3":
SELECT RECORD ContractionRow FROM ContractionTable
    WHERE field 1 matches SourceString[SourceIndex] and
    WHERE field 2 matches SourceString[SourceIndex + 1] and
    WHERE field 3 matches SourceString[SourceIndex + 2]

COMMENT If this sequence isn't a contraction then one will not be found
IF ContractionRow is null THEN
    SET Result to null
    RETURN false
ENDIF

COMMENT Found a contraction, get its weights
SET Result.ScriptMember to ContractionRow.Field4
SET Result.PrimaryWeight to ContractionRow.Field5

SET Result.DiacriticWeight to ContractionRow.Field6
SET Result.CaseWeight to ContractionRow.Field7

"2":
SELECT RECORD ContractionRow FROM ContractionTable
    WHERE field 1 matches SourceString[SourceIndex] and
    WHERE field 2 matches SourceString[SourceIndex + 1]

COMMENT If this sequence isn't a contraction then one will not be found
IF ContractionRow is null THEN
    SET Result to null
    RETURN false
ENDIF

COMMENT Found a contraction, get its weights
SET Result.ScriptMember to ContractionRow.Field3
SET Result.PrimaryWeight to ContractionRow.Field4

SET Result.DiacriticWeight to ContractionRow.Field5
SET Result.CaseWeight to ContractionRow.Field6

ENDCASE

SET UnicodeWeight to

```

```

    CorrectUnicodeWeight(Result, IsKoreanLocale)
    APPEND UnicodeWeight to UnicodeWeights
    APPEND Result.DiacriticWeight to DiacriticWeights as a BYTE
    APPEND Result.CaseWeight to CaseWeights as a BYTE

    COMMENT Advance the source index
    SET SourceIndex to SourceIndex + ContractionType

    RETURN true

```

3.1.5.2.13 Check3ByteWeightLocale

This algorithm checks if the specified locale is a CJK (Chinese/Japanese/Korean) sorting locale that uses third byte in Unicode weight.

```

COMMENT Check3ByteWeightLocale
COMMENT
COMMENT On Entry: SortLocale - Locale to use for linguistic sorting data
COMMENT
COMMENT On Exit: Result: Set to true if the specified locale is a CJK
COMMENT (Chinese/Japanese/Korean) locale that uses third byte in Unicode weight
COMMENT

SET Result to false

CASE SortLocale
    "0x0404": // Taiwan (Stroke Count)
    "0x0804": // China (Pronunciation)
    "0x0c04": // Hong Kong (Stroke Count)
    "0x1004": // Singapore (pronunciation)
    "0x1404": // Macau (pronunciation)
    "0x20804": // China (Stroke Count)
    "0x21004": // Singapore (Stroke Count)
    "0x21404": // Macau (Stroke Count)
    "0x30404": // Taiwan (Bopomofo)
    "0x40411": // Japanese (Radical / Stroke)
        SET Result to true
ENDCASE

RETURN Result

```

3.1.5.2.14 SpecialCaseHandler

This algorithm specifies the special processing that is required based on a different script member type.

```

COMMENT SpecialCaseHandler
COMMENT
COMMENT On Entry: SourceString - Source Unicode String
COMMENT SourceIndex - Current Index within source
COMMENT string
COMMENT UnicodeWeights - String of UnicodeWeightType to
COMMENT append additional weight(s) to
COMMENT ExtraWeights - String of ExtraWeightType to
COMMENT append extra weight(s) to if
COMMENT needed
COMMENT SpecialWeights - String of SpecialWeightType to
COMMENT append special weight(s) to
COMMENT if needed
COMMENT SortLocale - Locale to use for linguistic
COMMENT sorting data

```

```

COMMENT          IsKoreanLocale - True if this locale needs
COMMENT          Korean special casing of the
COMMENT          ScriptMember value
COMMENT On Exit:  SourceIndex    - Index of last character
COMMENT          processed, caller will need to
COMMENT          loop increment to continue
COMMENT          Korean Jamo cases can increment
COMMENT          this beyond its input value
COMMENT          UnicodeWeights - The UnicodeWeight of the
COMMENT          processed character(s) is
COMMENT          appended to this string.
COMMENT          ExtraWeights   - The ExtraWeight, if any, of
COMMENT          the processed character(s) is
COMMENT          appended to this string.
COMMENT          SpecialWeights - The Special Weight, if any,
COMMENT          of the processed character(s)
COMMENT          is appended to this string.
COMMENT

PROCEDURE SpecialCaseHandler (IN SourceString : Unicode String
INOUT SourceIndex : 32 bit integer
INOUT UnicodeWeights : UnicodeWeightType String,
INOUT ExtraWeights : ExtraWeightType String,
INOUT SpecialWeights : SpecialWeightType String,
IN SortLocale : LCID,
IN IsKoreanLocale : boolean)

// Get the weight for the current character
SET CharacterWeights to CALL GetCharacterWeights WITH
    (SortLocale, SourceString[SourceIndex])
CASE CharacterWeight.ScriptMember OF
    UNSORTABLE :
        // Character is unsortable, so skip it
        RETURN
    NONSPACE MARK :
        // Character is a nonspace mark, so only store the
        // diacritic weight.
        If (Length(DiacriticWeights) is greater than 0) THEN
            SET last DiacriticWeight in DiacriticWeights to
                DiacriticWeight + CharacterWeights.DiacriticWeight
        ELSE
            APPEND CharacterWeights.DiacriticWeight to DiacriticWeights as a BYTE
        ENDIF
        RETURN
    EXPANSION :
        // Expansion character, each character has 2 weights, store
        // each weight separately
        SET Weights to CALL GetExpansionWeights WITH
            (SourceString[SourceIndex], SortLocale)
        // Store the appropriate weights, 2 or 3
        FOR each Weight in Weights
            // Store the weight of the first character of the
            // expansion
            SET UnicodeWeight to CALL CorrectUnicodeWeight WITH
                (Weights, IsKoreanLocale)
            APPEND UnicodeWeight to UnicodeWeights
            APPEND Weights.DiacriticWeight to DiacriticWeights as a BYTE
            APPEND Weights.CaseWeight to CaseWeights as a BYTE
        ENDFOR
        RETURN
    PUNCTUATION :
        SET Position to Length(UnicodeWeights) as 16 bit integer
        APPEND Position into SpecialWeights as 16 bit integer
        SET SpecialWeight to CALL MakeUnicodeWeight WITH
            (CharacterWeight.ScriptMember,
            CharacterWeight.PrimaryWeight, False)
        APPEND SpecialWeight to SpecialWeights as 16 bit integer
        RETURN
    SYMBOL_1 :

```

```

SYMBOL_2 :
SYMBOL_3 :
SYMBOL_4 :
SYMBOL_5 :
SYMBOL_6 :
    // Character is a symbol, store Unicode Weights
    SET UnicodeWeight to CALL CorrectUnicodeWeight WITH
        (Weights[0], IsKoreanLocale)
    APPEND UnicodeWeight to UnicodeWeights
    APPEND CharacterWeights.DiacriticWeight to DiacriticWeights as a BYTE
    APPEND CharacterWeights.CaseWeight to CaseWeights as a BYTE
    RETURN
EASTASIA_SPECIAL :
    // Get the primary and case weight of the current code point
    SET PrimaryWeight to UnicodeWeight.PrimaryWeight
    SET ExtraWeight to UnicodeWeight.CaseWeight
    // Mask off the bits that are not required
    SET ExtraWeight to (ExtraWeight & CaseMask) |
        CASE EXTRA WEIGHT MASK
    // Special case Repeat and Cho-On
    // PrimaryWeight = 0 => Repeat
    // PrimaryWeight = 1 => Cho-On
    // PrimaryWeight = 2+ => Kana
    IF PrimaryWeight is less than or equal to MAX_SPECIAL_PW THEN
        // If the script member of the previous character is
        // invalid, then give the special character
        // invalid weight (highest possible weight) so that it
        // will sort AFTER everything else.
        SET PreviousIndex to SourceIndex - 1
        SET UnicodeWeight.ScriptMember to MAP_INVALID_WEIGHT
        SET UnicodeWeight.PrimaryWeight to MAP_INVALID_WEIGHT
        WHILE PreviousIndex is greater than or equal to 0
            SET PreviousWeight to CALL GetCharacterWeights WITH
                (SortLocale, SourceString[PreviousIndex])
            IF PreviousWeight.ScriptMember is less than
                EASTASIA_SPECIAL THEN
                IF PreviousWeight.ScriptMember is not equal to
                    EXPANSION THEN
                    // UNSORTABLE or NONSPACE_MARK
                    // Ignore these to get the
                    // previous ScriptMember/PrimaryWeight
                    DECREMENT PreviousIndex
                    CONTINUE WHILE PreviousIndex
                ENDIF
            ELSE IF PreviousWeight.ScriptMember is equal to
                EASTASIA_SPECIAL THEN
                IF PreviousWeight.PrimaryWeight is less than or equal to
                    MAX_SPECIAL_PW THEN
                    // Handle case where two special chars follow
                    // each other. Keep going back in the string
                    DECREMENT PreviousIndex
                    CONTINUE WHILE PreviousIndex
                ENDIF
            SET UnicodeWeight to
                CALL MakeUnicodeWeight WITH (KANA,
                    PreviousWeight.PrimaryWeight, IsKoreanLocale)
            // Only build weights W6 & W7 if the previous
            // character is KANA.
            // ignores W4 & W5
            // Always:
            // W6 = previous CW & ISOLATE_KANA
            SET PreviousExtraWeight to PreviousWeight.CaseWeight
            // Mask off the bits that aren't required
            SET PreviousExtraWeight to CASE_EXTRA_WEIGHT_MASK |
                (PreviousExtraWeight & CaseMask)
            // Ignore kana and width
            // so these are merely CASE_EXTRA_WEIGHT_MASK
            SET ExtraWeight.W6 to CASE_EXTRA_WEIGHT_MASK
            SET ExtraWeight.W7 to CASE_EXTRA_WEIGHT_MASK

```

```

        // Repeat is already done, which is:
        // UW = previous UW (set above)
        // W5 = ignored
        // W7 = previous CW & ISOLATE_WIDTH (done above)
        IF PrimaryWeight is not equal to PW_REPEAT THEN
            // Cho-On:
            // UW = previous UW & CHO_ON_UW_MASK
            // W5 = ignored
            // W7 = current CW & ISOLATE_WIDTH (done above)
            SET UnicodeWeight.PrimaryWeight to
            UnicodeWeight.PrimaryWeight & CHO ON PW MASK
        ENDIF
        // Append the calculated ExtraWeight
        // APPEND ExtraWeight to ExtraWeights
    ELSE
        // The previous weight is not EASTASIA_SPECIAL, so just
        // store the previous weight
        SET UnicodeWeight to CorrectUnicodeWeight
        (PreviousWeight, IsKoreanLocale)
        // Append the weight that was found
        APPEND UnicodeWeight to UnicodeWeights
    ENDIF
ENDWHILE
ELSE
    // Kana
    // ScriptMember = KANA
    // PrimaryWeight = current PrimaryWeight
    // W4 = current CaseWeight & ISOLATE_SMALL
    // W5 = WT FIVE KANA
    // W6 = current CaseWeight & ISOLATE_KANA
    // W7 = current CaseWeight & ISOLATE_WIDTH
    SET UnicodeWeight to CALL MakeUnicodeWeight WITH ( KANA,
        CharacterWeight.PrimaryWeight, IsKoreanLocale)
    APPEND UnicodeWeight to UnicodeWeights
    SET TempExtraWeight.W4 to ExtraWeight & ISOLATE_SMALL
    SET TempExtraWeight.W5 to WT FIVE KANA
    SET TempExtraWeight.W6 to ExtraWeight & ISOLATE_KANA
    SET TempExtraWeight.W7 to ExtraWeight & ISOLATE_WIDTH
    APPEND TempExtraWeight to ExtraWeights
ENDIF
APPEND CharacterWeight.DiacriticWeight to DiacriticWeights as a BYTE
APPEND MIN CW to CaseWeights as a BYTE
RETURN

JAMO_SPECIAL :
    // See if it's a leading Jamo
    IF (CALL IsJamoLeading(SourceString[SourceIndex])) is true
        THEN
            // If the characters beginning at SourceIndex are a valid
            // old Hangul composition, create the SortKey
            // according to the old Hangul rule
            SET OldHangulCount to
            CALL MapOldHangulSortKey WITH (SourceString,
                SourceIndex, SortLocale, UnicodeWeights, IsKoreanLocale)
            IF OldHangulCount is greater than 0 THEN
                // Decrement OldHangulCount because the caller's loop
                // will increment the SourceIndex as well
                DECREMENT OldHangulCount
                SET SourceIndex to SourceIndex + OldHangulCount
                RETURN
            ENDIF
        ENDIF
        // Otherwise, fall back to the normal behavior
        // No special case on the character, so store the Jamo's
        // weights.
        // Store the real script member in the diacritic weight
        // in the tables since both the diacritic weight and the
        // case weight are not used in Korean
        // For example, from unisort.txt:

```

```

        // 0x1101 4 84 83 2 ;   Choseong Ssangkiyeok
        // Field 2 has a value of 4 to trigger the code case for JAMO_SPECIAL.
        // Field 3 (84) is the real primary weight for this Jamo.
        // Field 4 (83) is the real script member for this Jamo.
        SET UnicodeWeight to CALL MakeUnicodeWeight WITH
            (CharacterWeight.DiacriticWeight,
             CharacterWeight.PrimaryWeight, IsKoreanLocale)
        APPEND UnicodeWeight to UnicodeWeights
        APPEND MIN_DW to DiacriticWeights as a BYTE
        APPEND MIN_CW to DiacriticWeights as a BYTE
        RETURN
EXTENSION A :
    // Extension A gives us two weights
    // UnicodeWeight = SM_EXT_A, AW_EXT_A, AW, DW
    // First Weight
    SET UnicodeWeight to CALL MakeUnicodeWeight WITH
        (SCRIPT MEMBER EXT A, PRIMARY WEIGHT EXT A,
         IsKoreanLocale)
    APPEND UnicodeWeight to UnicodeWeights
    // Since the script member is our flag for this EXTENSION_A special
    // case, the real weights are in fields 2 & 3.
    // Example:
    // From unisort.txt:
    // 0x3400 5 16 2 2 ; ㅏ CJK Unified Ideographs Extension A
    // Field 2 is the script member.
    // Field 3 is the primary weight.
    // Second Weight
    SET UnicodeWeight to CALL MakeUnicodeWeight WITH
        (CharacterWeight.PrimaryWeight,
         CharacterWeight.DiacriticWeight, false)
    APPEND UnicodeWeight to UnicodeWeights
    APPEND MIN_DW to DiacriticWeights as a BYTE
    APPEND MIN_CW to DiacriticWeights as a BYTE
    RETURN
ENDCASE

```

3.1.5.2.15 GetPositionSpecialWeight

This algorithm specifies the retrieval of special weight based on the source index.

```

COMMENT GetPositionSpecialWeight
COMMENT
COMMENT On Entry:  Position - Position to calculate weight for
COMMENT
COMMENT On Exit:   Weight    - Resulting weight
COMMENT

PROCEDURE GetPositionSpecialWeight(IN Position : 32 bit integer,
                                   OUT Weight : 16 bit integer)

    // Add some bits (0x8003) to adjust the weight and because
    // some bits are expected. Since setting 0x3 is required, rotate the source
    // index 2 bits so as to not lose the precision.

    // Note that if SourceIndex is larger than 0x1FFF, then some bits
    // will be lost on the conversion to 16 bits. Presumably if a string
    // is over 8191 characters long, they will differ well before this
    // point, so the lost information is irrelevant.

    SET Weight to (SourceIndex << 2) | 0x8003
    RETURN Weight

```

3.1.5.2.16 MapOldHangulSortKey

This algorithm specifies the generation of Unicode weight based on the strings at the specified index that have a special Old Hangul sequence. <5>

3.1.5.2.17 GetJamoComposition

This algorithm specifies the strings at the specified index that form a valid Old Hangul character that is composed of a Jamo character sequence. <6>

```
COMMENT GetJamoComposition
COMMENT
COMMENT On Entry:  SourceString - Unicode String to test
COMMENT              CurrentIndex - Index of leading Jamo to start from
COMMENT              JamoClass    - Class of Jamo to look for
COMMENT              JamoSortInfo - Information about the current
COMMENT                               sequence
COMMENT On Exit:   JamoSortInfo - Updated with information about
COMMENT                               the new sequence
COMMENT              SourceIndex - Updated to next character if
COMMENT                               Jamo is found
COMMENT              NewJamoClass - New class to look for next
COMMENT
COMMENT NOTE: This function assumes the character at SourceString
COMMENT         [SourceIndex] is a leading Jamo.
COMMENT         Ie: IsJamo() returned true
COMMENT

PROCEDURE GetJamoComposition (IN SourceString : Unicode String,
                              INOUT CurrentIndex : 32 bit integer,
                              IN JamoClass : enumeration,
                              INOUT JamoSortInfo : JamoSortInfoType,
                              OUT NewJamoClass : enumeration)

SET CurrentCharacter to SourceString[CurrentIndex]

// Get the Jamo information for the current character
SET JamoStateData to CALL GetJamoStateData WITH (CurrentCharacter)
SET JamoSortInfo to CALL UpdateJamoSortInfo
    WITH (JamoClass, JamoStateData, JamoSortInfo)

// Move on to the next character
INCREMENT CurrentIndex

WHILE CurrentIndex is less than Length(SourceString)
    SET CurrentCharacter to SourceString[CurrentIndex]

    IF CALL IsJamo WITH (CurrentCharacter) is not true THEN
        // The current character is not a Jamo,
        // Done checking for a Jamo composition
        SET NewJamoClass to "Invalid Jamo Sequence"
        RETURN
    ENDIF

    IF CurrentCharacter is equal to 0x1160 THEN
        SET JamoSortInfo.FillerUsed to true
    ENDIF

    // Get the Jamo class of it
    IF CALL IsJamoLeading WITH (CurrentCharacter) is true THEN
        SET NewJamoClass to "Leading Jamo Class"
    ELSE IF CALL IsJamoTrailing WITH (CurrentCharacter) is true THEN
        SET NewJamoClass to "Trailing Jamo Class"
    ELSE
        SET NewJamoClass to "Vowel Jamo Class"
    ENDIF

    IF JamoClass is not equal to NewJamoClass THEN
        RETURN NewJamoClass
    ENDIF
```

```

// Push the current Jamo (SourceString[CurrentIndex])
// into the state machine to check if it is a valid
// old Hangul composition. During the check also
// update the sortkey result in:
JamoSortInfo

// Find the new record
SET JamoStateData to CALL FindNewJamoState
    WITH (CurrentCharacter, JamoStateData)

// A valid old Hangul composition was not found for the current
// character so return the current Jamo class
// (JamoClass and NewJamoClass are identical)
IF JamoStateData is null THEN
    RETURN NewJamoClass
ENDIF

// A match has been found, so update our info.
SET JamoSortInfo to CALL UpdateJamoSortInfo
    WITH (JamoClass, JamoStateData, JamoSortInfo)

// Still in a valid old Hangul composition.
//Go check the next character.
INCREMENT CurrentIndex

ENDWHILE CurrentIndex

SET NewJamoClass to "Invalid Jamo Sequence"
RETURN NewJamoClass

```

3.1.5.2.18 GetJamoStateData

This algorithm specifies the retrieval of state machine information to check if the specified Jamo sequence forms a valid Old Hangul character. [<7>](#)

3.1.5.2.19 FindNewJamoState

This algorithm specifies retrieval of a new state from the state machine for Jamo processing. [<8>](#)

```

COMMENT FindNewJamoState
COMMENT
COMMENT On Entry:  JamoCharacter    - Unicode Character to get Jamo
COMMENT                                     information for
COMMENT                JamoStateData - Current Jamo state information
COMMENT
COMMENT On Exit:   JamoStateData    - New Jamo state record from the
COMMENT                                     data file, null if an
COMMENT                                     appropriate state record is
COMMENT                                     not found.
COMMENT

PROCEDURE FindNewJamoState(IN JamoCharacter : Unicode Character,
                          INOUT JamoStateData : JamoStateDataType)

// The current JamoStateData.DataRecord points to the base record.
// There are JamoStateData.TransitionCount following records that can
// match the input JamoCharacter, the search is for the first one
SET DataRecord to JamoStateData.DataRecord

WHILE JamoStateData.TransitionCount is greater than 0
    // advance to the next record in the data and test if
    // it is the correct record for JamoCharacter
    ADVANCE DataRecord to next record in data table
    IF DataRecord.Field1 is equal to JamoCharacter THEN

```

```

        // Found a record, get its info and return it
        // Now gather the information from that record.
        SET JamoStateData.OldHangulFlag    to JamoRecord.Field2
        SET JamoStateData.LeadingIndex     to JamoRecord.Field3
        SET JamoStateData.VowelIndex       to JamoRecord.Field4
        SET JamoStateData.TrailingIndex    to JamoRecord.Field5
        SET JamoStateData.ExtraWeight      to JamoRecord.Field6
        SET JamoStateData.TransitionCount to JamoRecord.Field7

        // Remember the record
        SET JamoStateData.DataRecord to JamoRecord

        RETURN JamoStateData
    ENDWHILE

    // record not found, return null
    SET JamoStateData to null
    RETURN JamoStateData

```

3.1.5.2.20 UpdateJamoSortInfo

This algorithm specifies the update of Jamo sorting information based on the current state of the state machine for Jamo processing. [<9>](#9)

3.1.5.2.21 IsJamo

This algorithm specifies the check for a valid Jamo character. [<10>](#10)

```

COMMENT IsJamo
COMMENT
COMMENT On Entry:  SourceCharacter - Unicode Character to test
COMMENT
COMMENT On Exit:   Result           - true if SourceCharacter is in
COMMENT                               the Jamo range
COMMENT
PROCEDURE IsJamoLeading(IN SourceCharacter : Unicode Character,
                      OUT Result: boolean)

IF (SourceCharacter is greater than or equal to NLS_CHAR_FIRST_JAMO)
    and
    (SourceCharacter is less than or equal to NLS_CHAR_LAST_JAMO) THEN
    SET Result to true
ELSE
    SET Result to false
ENDIF

RETURN Result

```

3.1.5.2.22 IsCombiningJamo

This algorithm specifies the check for a valid Jamo character. [<11>](#11)

```

COMMENT IsCombiningJamo
COMMENT
COMMENT On Entry:  SourceCharacter - Unicode Character to test
COMMENT
COMMENT On Exit:   Result           - true if SourceCharacter is in
COMMENT                               the Jamo range
COMMENT
PROCEDURE IsJamoLeading(IN SourceCharacter : Unicode Character,

```

```

OUT Result: boolean)

IF ((SourceCharacter is greater than or equal to NLS_CHAR_FIRST_JAMO)
    and
    (SourceCharacter is less than or equal to NLS_CHAR_LAST_JAMO))
    Or
    ((SourceCharacter is greater than or equal to NLS_CHAR_FIRST_EXT_A_LEADING_JAMO)
    and
    (SourceCharacter is less than or equal to NLS_CHAR_LAST_EXT_A_LEADING_JAMO))
    Or
    ((SourceCharacter is greater than or equal to NLS_CHAR_FIRST_EXT_B_VOWEL_JAMO)
    and
    (SourceCharacter is less than or equal to NLS_CHAR_LAST_EXT_B_VOWEL_JAMO))
    Or
    ((SourceCharacter is greater than or equal to NLS_CHAR_FIRST_EXT_B_TRAILING_JAMO)
    and
    (SourceCharacter is less than or equal to NLS_CHAR_LAST_EXT_B_TRAILING_JAMO)) THEN
    SET Result to true
ELSE
    SET Result to false
ENDIF

RETURN Result

```

3.1.5.2.23 IsJamoLeading

This algorithm checks if the specified Jamo character is a leading Jamo. [<12>](#<12>)

3.1.5.2.24 IsJamoVowel

This algorithm checks whether the specified Jamo character is a vowel Jamo. [<13>](#<13>)

```

COMMENT IsJamoVowel
COMMENT
COMMENT On Entry: SourceCharacter - Unicode Character to test
COMMENT
COMMENT On Exit: Result - true if this is a vowel Jamo
COMMENT

PROCEDURE IsJamoTrailing(IN SourceCharacter : Unicode Character,
    OUT Result: boolean)

IF ((SourceCharacter is greater than or equal to NLS_CHAR_FIRST_VOWEL_JAMO)
    and
    (SourceCharacter is less than or equal to NLS_CHAR_LAST_VOWEL_JAMO))
    Or
    ((SourceCharacter is greater than or equal to NLS_CHAR_FIRST_EXT_B_VOWEL_JAMO)
    and
    (SourceCharacter is less than or equal to NLS_CHAR_LAST_LEADING_EXT_B_VOWEL_JAMO))
    SET Result to true
ELSE
    SET Result to false
ENDIF

RETURN Result

```

3.1.5.2.25 IsJamoTrailing

This algorithm checks if the specified Jamo character is a trailing Jamo. [<14>](#<14>)

```

COMMENT IsJamoTrailing
COMMENT

```

```

COMMENT On Entry: SourceCharacter - Unicode Character to test
COMMENT
COMMENT On Exit: Result - true if this is a trailing Jamo
COMMENT
COMMENT NOTE: Only call this if the character is known to be a Jamo
COMMENT syllable. This function only helps distinguish between
COMMENT the different types of Jamo, so only call it if
COMMENT IsJamo() has returned true.
COMMENT

PROCEDURE IsJamoTrailing(IN SourceCharacter : Unicode Character,
                        OUT Result: boolean)

IF SourceCharacter is greater than
  or equal to NLS_CHAR_FIRST_VOWEL_JAMO THEN
  SET Result to true
ELSE
  SET Result to false
ENDIF

RETURN Result

```

3.1.5.2.26 InitKoreanScriptMap

This algorithm specifies the initialization of a data structure that is required for the special processing of Korean script members.

```

COMMENT InitKoreanScriptMap
COMMENT
COMMENT On Entry: global KoreanScriptMap - presumed to be null
COMMENT
COMMENT On Exit: global KoreanScriptMap - initialized to map
COMMENT scripts to Korean
COMMENT
COMMENT This procedure initializes the Korean, causing ideographic
COMMENT scripts to sort prior to other scripts for the Korean.
COMMENT

PROCEDURE InitKoreanScriptMap

SET KoreanScriptMap to new array of 256 null bytes

// Initialize the "scripts" prior to first script (Latin, script 14)
FOR counter is 0 to FIRST_SCRIPT - 1
  SET KoreanScriptMap[counter] to counter
ENDFOR counter

// For Korean the Ideographs sort to the first script,
// so start with that index
SET NewScript to FIRST_SCRIPT

// Test if the IDEOGRAPH script is part of a multiple weights script

// For convenience hard code the information from the
// unisort.txt section SORTTABLES\MULTIPLEWEIGHTS
// IDEOGRAPHS are 128 through 241,
// map them to FIRST_SCRIPT through 127
FOR counter is IDEOGRAPH to 241
  SET KoreanScriptMap[counter] to NewScript
  INCREMENT NewScript
ENDFOR

// Now set the remaining unset scripts the next NewScript value
FOR counter is 0 to MAX_SCRIPTS - 1
  // If the value has not been set yet, set it to the next value
  IF KoreanScriptMap[counter] is null THEN

```

```

        SET KoreanScriptMap[counter] to NewScript
        INCREMENT NewScript
    ENDIF
ENDFOR

```

3.1.5.3 Mapping UTF-16 Strings to Upper Case

To map a UTF-16 string to upper case, each UTF-16 code point is looked for in an upper casing table [\[MSDN-UCMT/Win8\]](#). If an entry is found, the input code point is changed to the output code point.

3.1.5.3.1 ToUpperCase

This algorithm converts a UTF-16 string to its upper case form.

```

COMMENT ToUpperCase
COMMENT On Entry: inputString - A string encoded in UTF-16
COMMENT
COMMENT On Exit: Result      - A string encoded in UTF-16 with
COMMENT                      the output in Upper Case form.

PROCEDURE ToUpperCase

SET Result to empty string

SET index to 0
WHILE index is less than Length(inputString)
    SET upperCase to UpperCaseMapping(inputString[index])
    APPEND upperCase to Result
    INCREMENT index
ENDWHILE

RETURN

```

3.1.5.3.2 UpperCaseMapping

This algorithm converts a UTF-16 code point to its upper case form using the UpperCaseTable in [\[MSDN-UCMT/Win8\]](#).

```

COMMENT UpperCaseMapping
COMMENT On Entry: SourceCharacter - A UTF-16 code point
COMMENT
COMMENT On Exit: Result      - Upper case UTF-16 code point

PROCEDURE UpperCaseMapping

SELECT RECORD caseMapping FROM UpperCaseTable WHERE field 1
    matches SourceCharacter
IF EXISTS caseMapping
    SET Result TO caseMapping field 2
ELSE
    SET Result TO SourceCharacter
ENDIF

RETURN

```

3.1.5.4 Unicode International Domain Names

International Domain Name support is provided by IdnToNameprepUnicode, IdnToAscii, and IdnToUnicode. The algorithms follow either the **IDNA2003** or **IDNA2008+UTS46** standards depending on the specific implementation environment. [<15>](#)

3.1.5.4.1 IdnToAscii

```
COMMENT IdnToAscii
COMMENT On Entry:  SourceString - Unicode String to get Punycode
COMMENT                               representation of.
COMMENT                               Flags      - Bit flags to control behavior
COMMENT                               of IDN validation
COMMENT
COMMENT IDN_ALLOW_UNASSIGNED:    During validation, allow unicode
COMMENT                               code points that are not assigned.
COMMENT IDN_USE_STD3_ASCII_RULES: Enforce validation of the STD3
COMMENT                               characters.
COMMENT IDN_EMAIL_ADDRESS:      Allow punycode encoding of the local part
COMMENT                               of an email address to tunnel EAI
COMMENT                               addresses through non-Unicode slots.
COMMENT
COMMENT On Exit:  Punycode      - String containing the Punycode ASCII range
COMMENT                               form of the input
PROCEDURE IdnToAscii(IN SourceString : Unicode String,
                    IN Flags: 32 bit integer,
                    OUT PunycodeString : Unicode String)

COMMENT Split input string into email local part and domain parts
COMMENT as appropriate
IF (IDN_EMAILADDRESS bit is on in Flags) THEN
    IF (SourceString CONTAINS "@") THEN
        SET arrayParts = SourceString.Split("@")
        SET emailLocalString to arrayParts[0]
        SET domainString to arrayParts[1]
    ELSE
        SET emailLocalString to SourceString
        SET domainString to ""
    ENDIF
ELSE
    SET domainString to SourceString
    SET emailLocalString to ""
ENDIF

SET OutputString TO ""

IF (emailLocalString IS NOT EMPTY) THEN
    COMMENT email local part cannot contain null character
    IF (emailLocalString CONTAINS character U+0000) THEN
        RETURN ERROR
    ENDIF

    COMMENT email local part is normalized per Normalization Form C (NFC)
    COMMENT Defined in Unicode Technical Report #15 (UTR#15)
    COMMENT http://www.unicode.org/reports/tr15/tr15-18.html
    ApplyUTR15NormalizationFormC(emailLocalString)

    IF (emailLocalString CONTAINS character U+0080 through character U+10FFFF) THEN
        encodedString = PunycodeEncode(emailLocalString)
        PREPEND "x1--" TO encodedString
    ELSE
        SET encodedString TO emailLocalString
    ENDIF

    COMMENT email local part cannot be > 255 characters even converted
    IF (LENGTH of encodedString IS GREATER THAN 255) THEN
        RETURN ERROR
    ENDIF
```

```

    SET OutputString TO encodedString

    COMMENT Will need an @ if there is a domain part too
    IF (domainString IS NOT EMPTY) THEN
        APPEND "@" TO domainString
    ENDIF
ELSE
    COMMENT Cannot have empty local part in email mode
    IF (IDN_EMAIL_ADDRESS bit is on in Flags) THEN
        RETURN ERROR
    ENDIF
ENDIF

IF (domainString IS NOT EMPTY) THEN
    (domainString is not empty) THEN

    COMMENT See if STD3 rules need tested
    COMMENT Test for invalid characters in domain name
    IF ((IDN_USE_STD3_ASCII_RULES bit is on in Flags) AND
        ((domainString CONTAINS characters U+0000 through ',') OR
         (domainString CONTAINS character '/') OR
         (domainString CONTAINS characters ':' through '@') OR
         (domainString CONTAINS characters '[' through '`') OR
         (domainString CONTAINS characters '{' through U+007F))) THEN
        RETURN ERROR
    ENDIF

    COMMENT Each Label of the domain name is processed independently
    DEFINE domainString AS Array OF String
    IF (domainString CONTAINS ".") THEN
        SET domainLabels TO domainString.Split(".")
    ELSE
        SET domainLabels[0] TO domainString
    ENDIF

    SET encodedDomain TO ""

    FOREACH label IN domainLabels DO

        SET encodedString TO ""
        IF (label CONTAINS characters U+0080 THROUGH U+10FFFF) THEN
            IF Windows version is Windows Vista, Windows Server 2008, Windows 7, or
            Windows Server 2008 R2 THEN
                SET normalizedLabel TO NormalizeForIdna2003(label, flags)
            ELSE
                SET normalizedLabel TO NormalizeForIdna2008(label, flags)
            ENDIF

            SET encodedString TO PunycodeEncode(normalizedLabel)
            PREPEND "xn--" TO encodedString

        ELSE
            COMMENT ASCII range only, does not need encoding
            SET encodedString TO label
        ENDIF

    COMMENT domain labels cannot be empty or > 63 characters even converted
    IF ((LENGTH OF encodedString IS EMPTY) OR
        (LENGTH OF encodedString IS GREATER THAN 63)) THEN
        RETURN ERROR
    ENDIF

    COMMENT See if STD3 rules need tested
    IF (IDN_USE_STD3_ASCII_RULES bit is on in Flags)
        COMMENT domain labels cannot be empty
        IF (label IS EMPTY) THEN
            RETURN ERROR
        ENDIF

```

```

        COMMENT leading and trailing - are illegal in domain labels
        IF (label BEGINS WITH "-" OR
            label END WITH "-") THEN
            RETURN ERROR
        ENDIF
    ENDIF

    COMMENT Need to retain separators between domain labels
    IF (label IS NOT LAST VALUE IN domainLabels) THEN
        APPEND "." to encodedDomain
    ENDIF

ENDFOREACH

COMMENT encoded domains cannot be > 255 characters.
IF (LENGTH OF encodedDomain IS GREATER THAN 255) THEN
    RETURN ERROR
ENDIF

APPEND encodedDomain to OutputString
ENDIF

RETURN OutputString

```

3.1.5.4.2 IdnToUnicode

```

COMMENT IdnToUnicode
COMMENT On Entry: SourceString - Idn String to get Unicode
COMMENT                               representation of.
COMMENT                               Flags - Bit flags to control behavior
COMMENT                               of IDN validation
COMMENT
COMMENT IDN ALLOW UNASSIGNED: During validation, allow unicode
COMMENT                               code points that are not assigned.
COMMENT IDN USE STD3 ASCII RULES: Enforce validation of the STD3
COMMENT                               characters.
COMMENT IDN_RAW_PUNYCODE: Only decode the punycode, no additional
COMMENT                               validation.
COMMENT IDN EMAIL ADDRESS: Allow punycode encoding of the local part
COMMENT                               of an email address to tunnel EAI
COMMENT                               addresses through non-Unicode slots.
COMMENT
COMMENT On Exit: UnicodeString - String containing the Unicode form of the
COMMENT                               input string.
PROCEDURE IdnToUnicode (IN SourceString : Punycode String,
                        IN Flags: 32 bit integer,
                        OUT UnicodeString : Unicode String)
UnicodeString = PunycodeDecode(SourceString)

COMMENT IDN RAW PUNYCODE stops here
IF (IDN_RAW_PUNYCODE bit is on in Flags) THEN
    return UnicodeString
ENDIF
COMMENT Otherwise verify that the result round trips
RoundTripPunycodeString = IdnToAscii(UnicodeString, Flags)
IF (RoundTripPunycodeString IS NOT EQUAL TO UnicodeString)
    return ERROR
ENDIF
return UnicodeString

```

3.1.5.4.3 IdnToNameprepUnicode

This function merely returns the output of what IdnToUnicode(IdnToAscii(InputString)) would return.

```

COMMENT IdnToNameprepUnicode
COMMENT On Entry: SourceString - Unicode String to get nameprep form of
COMMENT                      Flags - Bit flags to control behavior
COMMENT                      of IDN validation
COMMENT
COMMENT IDN_ALLOW_UNASSIGNED: During validation, allow unicode
COMMENT                      code points that are not assigned.
COMMENT IDN_USE_STD3_ASCII_RULES: Enforce validation of the STD3
COMMENT                      characters.
COMMENT IDN_EMAIL_ADDRESS: Allow punycode encoding of the local part
COMMENT                      of an email address to tunnel EAI
COMMENT                      addresses through non-Unicode slots.
COMMENT
COMMENT On Exit: NameprepString -String containing the nameprep form of the
COMMENT                      input string.
PROCEDURE IdnToNameprepUnicode(IN SourceString : Punycode String,
                              IN Flags: 32 bit integer,
                              OUT UnicodeString : Unicode String)
SET AsciiString TO IdnToAscii(SourceString, Flags)
SET NameprepString TO IdnToUnicode(AsciiString, Flags)

return NameprepString

```

3.1.5.4.4 PunycodeEncode

PunycodeEncode encodes an input ASCII/Unicode string. If the input contains non-ASCII parts, then punycoded strings are output, prefixed with the **xn--** or **xl--** labels.

```

PROCEDURE PunycodeEncode(IN UnicodeString : Unicode String,
                        IN Flags: 32 bit integer,
                        OUT PunycodeString : Unicode String)

COMMENT Split input string into email local part and domain parts
IF (IDN_EMAILADDRESS bit is on in Flags) THEN
  IF (UnicodeString CONTAINS "@") THEN
    SET arrayParts = UnicodeString.Split("@")
    SET emailLocalString TO arrayParts[0]
    SET domainString TO arrayParts[1]
  ELSE
    SET emailLocalString TO UnicodeString
    SET domainString TO ""
  ENDIF
ELSE
  SET domainString TO PunycodeString
  SET emailLocalString TO ""
ENDIF

SET PunycodeString TO ""

IF (emailLocalString IS NOT "") THEN
  IF (emailLocalString CONTAINS U+0080 THROUGH U+10FFFF) THEN
    SET PunycodeString TO "xl--"

    COMMENT punycode_encode is described in RFC 3492
    COMMENT http://tools.ietf.org/html/rfc3492

    SET encodedString TO punycode_encode(emailLocalString)
    APPEND encodedString to PunycodeString
  ELSE
    COMMENT Local part of email was not encoded
    SET PunycodeString TO emailLocalString
  ENDIF
ENDIF

```

```

IF (domainString IS NOT "") THEN
  IF emailLocalString IS NOT "" THEN
    APPEND "@" TO PunycodeString
  ENDIF

  COMMENT Each Label of the domain name is parsed independently
  DEFINE domainString AS Array OF String
  IF (domainString CONTAINS ".") THEN
    SET domainLabels TO domainString.Split(".")
  ELSE
    SET domainLabels[0] TO domainString
  ENDIF

  FOREACH label IN domainLabels DO
    IF (label CONTAINS U+0080 THROUGH U+10FFFF) THEN
      COMMENT punycode_encode is described in RFC 3492
      COMMENT http://tools.ietf.org/html/rfc3492

      SET encodedLabel TO punycode_encode(label)
      PREPEND "xn--" TO encodedLabel
    ELSE
      SET encodedLabel TO label
    ENDIF

    APPEND encodedLabel TO PunycodeString

    COMMENT Need to retain separators between domain labels
    IF (label IS NOT LAST VALUE IN domainLabels) THEN
      APPEND "." TO PunycodeString
    ENDIF
  ENDFOREACH
ENDIF

return PunycodeString

```

3.1.5.4.5 PunycodeDecode

PunycodeDecode decodes an input all-ASCII string. If the input contains the xn-- or xl-- prefix the decoding algorithm is applied.

```

PROCEDURE PunycodeDecode(IN PunycodeString : Unicode String,
                        IN Flags: 32 bit integer,
                        OUT UnicodeString : Unicode String)

COMMENT Non-ASCII data is unexpected
IF (PunycodeString CONTAINS U+0080 through U+10FFFF) THEN
  Return ERROR
ENDIF

COMMENT Split input string into email local part and domain parts
IF (IDN_EMAILADDRESS bit is on in Flags) THEN
  IF (SourceString CONTAINS "@") THEN
    SET arrayParts = PunycodeString.Split("@")
    SET emailLocalString TO arrayParts[0]
    SET domainString TO arrayParts[1]
  ELSE
    SET emailLocalString TO PunycodeString
    SET domainString to ""
  ENDIF
ELSE
  SET domainString TO PunycodeString
  SET emailLocalString TO ""
ENDIF

SET UnicodeString TO ""

```

```

IF (emailLocalString IS NOT "") THEN
    IF (emailLocalString BEGINS WITH "xl-") THEN
        TRIM "xl--" FROM BEGINNING OF emailLocalString

        COMMENT punycode decode is described in RFC 3492
        COMMENT http://tools.ietf.org/html/rfc3492

        UnicodeString = punycode_decode(emailLocalString)
    ELSE
        COMMENT Local part of email was not encoded
        UnicodeString = emailLocalString
    ENDIF
ENDIF

IF (domainString IS NOT "") THEN
    IF emailLocalString IS NOT "" THEN
        APPEND "@" TO UnicodeString
    ENDIF

    COMMENT Each Label of the domain name is parsed independently
    DEFINE domainString as Array of String
    IF (domainString CONTAINS ".") THEN
        SET domainLabels TO domainString.Split(".")
    ELSE
        SET domainLabels[0] TO domainString
    ENDIF

    FOREACH label IN domainLabels DO
        IF (label BEGINS WITH "xn--") THEN
            TRIM "xn--" FROM BEGINNING OF label

            COMMENT punycode_decode is described in RFC 3492
            COMMENT http://tools.ietf.org/html/rfc3492

            SET decodedLabel TO punycode_decode(label)
        ELSE
            SET decodedLabel TO label
        ENDIF

        APPEND decodedLabel TO UnicodeString

        COMMENT Need to retain separators between domain labels
        IF (label IS NOT LAST VALUE IN domainLabels) THEN
            APPEND "." TO UnicodeString
        ENDIF
    ENDFOREACH
ENDIF

return UnicodeString

```

3.1.5.4.6 IDNA2008+UTS46 NormalizeForIdna

NormalizeForIdna prepares the input string for encoding, using the mapping/normalization rules provided by **IDNA2008+UTS46** (**IDNA2008** with [\[TR46\]](#) applied).<16>

```

COMMENT NormalizeForIdna2008
COMMENT On Entry: SourceString - Unicode String to prepare for IDNA
COMMENT Flags - Bit flags to control behavior
COMMENT of IDN validation
COMMENT
COMMENT IDN_ALLOW_UNASSIGNED: During validation, allow unicode
COMMENT code points that are not assigned.
COMMENT
COMMENT On Exit: Punycode - String containing the Punycode ASCII range
COMMENT form of the input

```

```

PROCEDURE NormalizeForIdna2008 (IN SourceString : Unicode String,
                                IN Flags: 32 bit integer,
                                OUT OutputString : Unicode String)
COMMENT Mapping is done per the tables published by Unicode by following
COMMENT RFC5892 as modified by UTS#46 section 2 "Unicode IDNA Compatibility Processing"
COMMENT Appendix A of RFC5892 is NOT applied.
COMMENT Effectively this mapping is merely applying the latest IdnaMappingTable.txt
COMMENT mappings, including the "deviation" mappings from http://www.unicode.org/Public/idna/
COMMENT
COMMENT Apply UTS#46 Section 4 steps 1 & 2 to the string with the "Transitional Processing"
COMMENT option for the four "deviation" characters. Steps 3 and 4 are done by the caller.
COMMENT http://www.unicode.org/reports/tr46/#Processing
OPEN mapping FILE "http://www.unicode.org/Public/idna/6.3.0/IdnaMappingTable.txt"
SET OutputString TO ""
FOREACH character IN SourceString
    FIND RECORD data IN mapping WHERE LINE CONTAINS character
    IF (data IS EMPTY) THEN
        IF (IDN_ALLOW_UNASSIGNED bit IS NOT ON in Flags) THEN
            RETURN ERROR
        ELSE
            APPEND character TO OutputString
        ENDIF
    ELSE
        SWITCH (data FIELD statusValue)
            CASE "valid"
            CASE "disallowed_STD3_valid"
                BREAK
            CASE "ignored"
                SET character TO ""
                BREAK
            CASE "mapped"
            CASE "disallowed_STD3_valid"
            CASE "deviation"
                SET character TO data FIELD mappingValue
                BREAK
        ENDSWITCH
        APPEND character TO OutputString
    ENDIF
ENDFOREACH
RETURN OutputString

```

3.1.5.4.7 IDNA2003 NormalizeForIdna

NormalizeForIdna prepares the input string for encoding, using the mapping/normalization rules provided by **IDNA2003**.[<17>](#)

```

COMMENT NormalizeForIdna2003
COMMENT On Entry: SourceString - Unicode String to prepare for IDNA
COMMENT Flags - Bit flags to control behavior
COMMENT of IDN validation
COMMENT
COMMENT IDN ALLOW UNASSIGNED: During validation, allow unicode
COMMENT code points that are not assigned.
COMMENT
COMMENT On Exit: Punycode - String containing the Punycode ASCII range
COMMENT form of the input
PROCEDURE NormalizeForIdna2003 (IN SourceString : Unicode String,
                                IN Flags: 32 bit integer,
                                OUT OutputString : Unicode String)

COMMENT Behavior is identical to the results of RFC 3491 (http://tools.ietf.org/html/rfc3491
)
COMMENT Make sure to allow unassigned code points if IDN ALLOW UNASSIGNED bit is set in Flags
SET OutputString TO ApplyRfc3491(SourceString, Flags)

```

```
RETURN OutputString
```

3.1.5.5 Comparing UTF-16 Strings Ordinally

To do a case-sensitive ordinal comparison of strings, a binary comparison of the **UTF-16** code points of the strings is done. To do a case-insensitive ordinal string comparison, [ToUpperCase](#) (section 3.1.5.3.1) is done on each string before doing the ordinal comparison.

3.1.5.5.1 CompareStringOrdinal Algorithm

This algorithm compares two **UTF-16** strings by doing an ordinal (binary) comparison. Optionally, the caller can request that the comparison be done on the uppercase form of the string.

```
COMMENT CompareStringOrdinal
COMMENT On Entry: StringA      - A UTF-16 string to be compared
COMMENT On Entry: StringB      - Second UTF-16 string to compare
COMMENT On Entry: IgnoreCaseFlag - TRUE to ignore case when comparing
COMMENT
COMMENT On Exit: Result - A value indicating if StringA is less than,
COMMENT                      equal to, or greater than StringB

PROCEDURE CompareStringOrdinal

IF IgnoreCaseFlag is TRUE THEN
    SET StringA TO ToUpperCase(StringA)
    SET StringB TO ToUpperCase(StringB)
ENDIF

SET index TO 0

WHILE index is less than Length(StringA) and
      index is also less than Length(StringB)

    IF StringA[index] is less than StringB[index] THEN
        SET Result TO "StringA is less than StringB"
        RETURN
    ENDIF
    IF StringA[index] is greater than StringB[index] THEN
        SET Result TO "StringA is greater than StringB"
        RETURN
    ENDIF
    INCREMENT index
ENDWHILE

IF Length(StringA) is equal to Length(StringB) THEN
    SET Result TO "StringA is equal to StringB"
ELSE IF Length(StringA) is less than Length(StringB) THEN
    SET Result TO "StringA is less than StringB"
ELSE
    Assert Length(StringA) has to be greater than Length(StringB)
    SET Result TO "StringA is greater than StringB"
ENDIF
RETURN
```

3.1.6 Timer Events

None.

3.1.7 Other Local Events

None.

4 Protocol Examples

None.

5 Security

The following sections specify security considerations for implementers of the Windows Protocols Unicode Reference.

5.1 Security Considerations for Implementers

None.

5.2 Index of Security Parameters

None.

6 Appendix A: Product Behavior

The information in this specification is applicable to the following Microsoft products or supplemental software. References to product versions include released service packs.

- Windows NT operating system
- Windows 2000 operating system
- Windows XP operating system
- Windows Server 2003 operating system
- Windows Vista operating system
- Windows Server 2008 operating system
- Windows 7 operating system
- Windows Server 2008 R2 operating system
- Windows 8 operating system
- Windows Server 2012 operating system
- Windows 8.1 operating system
- Windows Server 2012 R2 operating system
- Windows 10 operating system
- Windows Server 2016 operating system
- Windows Server operating system
- Windows Server 2019 operating system
- Windows Server 2022 operating system
- Windows 11 operating system
- Windows Server 2025 operating system

Exceptions, if any, are noted in this section. If an update version, service pack or Knowledge Base (KB) number appears with a product name, the behavior changed in that update. The new behavior also applies to subsequent updates unless otherwise specified. If a product edition appears with the product version, behavior is different in that product edition.

Unless otherwise specified, any statement of optional behavior in this specification that is prescribed using the terms "SHOULD" or "SHOULD NOT" implies product behavior in accordance with the SHOULD or SHOULD NOT prescription. Unless otherwise specified, the term "MAY" implies that the product does not follow the prescription.

[<1> Section 3.1.5.2.3](#): Only Windows NT, Windows 2000, Windows XP, Windows Server 2003, Windows Vista, Windows Server 2008, Windows 7, and Windows Server 2008 R2 use record count for DEFAULT.

[<2> Section 3.1.5.2.3](#): An LCID is used in Windows NT, Windows 2000, Windows XP, Windows Server 2003, Windows Vista, Windows Server 2008, Windows 7, and Windows Server 2008 R2.

[<3> Section 3.1.5.2.3.1](#): The files in the download map to specific Windows versions as follows:

Version	File Name
Windows NT 4.0 operating system, Windows 2000, Windows XP, and Windows Server 2003	Windows NT 4.0 through Windows Server 2003 Sorting Weight Table.txt
Windows Vista	Windows Vista Sorting Weight Table.txt
Windows Server 2008	Windows Server 2008 Sorting Weight Table.txt
Windows 7 and Windows Server 2008 R2	Windows 7 and Windows Server 2008 R2 Sorting Weight Table.txt
Windows 8, Windows 8.1, Windows Server 2012, and Windows Server 2012 R2	Windows 8 and Windows Server 2012 Sorting Weight Table.txt Windows 8 Upper Case Mapping Table.txt
Windows 10, Windows Server 2016, Windows Server operating system, and Windows Server 2019	Windows 10 Sorting Weight Table.txt
Windows Server 2022	Windows Server 2022 Sorting Weight Table.txt
Windows 11	Windows 11 Sorting Weight Table.txt

[<4> Section 3.1.5.2.4](#): The following algorithm for generation of sort keys for a specific UTF-16 string is only used in Windows NT, Windows 2000, Windows XP, Windows Server 2003, Windows Vista, Windows Server 2008, Windows 7, Windows Server 2008 R2, Windows 8, Windows 8.1, Windows Server 2012, Windows Server 2012 R2, Windows 10, Windows Server 2016, Windows Server operating system, Windows Server 2019, and Windows Server 2022.

```

STRUCTURE CharacterWeightType
(
    ScriptMember:      8 bit integer
    PrimaryWeight:     8 bit integer
    DiacriticWeight:   8 bit integer
    CaseWeight:        8 bit integer
)

STRUCTURE UnicodeWeightType
(
    ScriptMember:      8 bit integer
    PrimaryWeight:     8 bit integer
    ThirdByteWeight:   8 bit integer
)

STRUCTURE SpecialWeightType
(
    Position:          16 bit integer
    ScriptMember:      8 bit integer
    PrimaryWeight:     8 bit integer
)

STRUCTURE ExtraWeightType
(
    W6:                8 bit integer
    W7:                8 bit integer
)

SET constant LCID_KOREAN to 0x0412
SET constant LCID_KOREAN_UNICODE_SORT to 0x010412
SET constant LCID_HUNGARIAN to 0x040e

SET constant SORTKEY_SEPARATOR to 0x01
SET constant SORTKEY_TERMINATOR to 0x00

```

```

SET global KoreanScriptMap to InitKoreanScriptMap

//
// Script Member Values.
//
SET constant UNSORTABLE to 0
SET constant NONSPACE_MARK to 1
SET constant EXPANSION to 2
SET constant EASTASIA_SPECIAL to 3
SET constant JAMO_SPECIAL to 4
SET constant EXTENSION A to 5
SET constant PUNCTUATION to 6

SET constant SYMBOL_1 to 7
SET constant SYMBOL_2 to 8
SET constant SYMBOL_3 to 9
SET constant SYMBOL_4 to 10
SET constant SYMBOL_5 to 11
SET constant SYMBOL_6 to 12

SET constant DIGIT to 13

SET constant LATIN to 14
SET constant KANA to 34
SET constant IDEOGRAPH to 128

IF Windows version is Windows Vista, Windows Server 2008, Windows 7, or
Windows Server 2008 R2 THEN
SET constant MAX_SPECIAL_CASE to SYMBOL_6
ELSE
SET constant MAX_SPECIAL_CASE to SYMBOL_5
ENDIF
COMMENT Set the constant for the first script member of the Unicode
COMMENT Private Use Area (PUA) range
SET constant PUA3BYTESTART to 0xA9
COMMENT Set the constant for the last script member of the Unicode
COMMENT Private Use Area (PUA) range
SET constant PUA3BYTEEND to 0xAF

COMMENT Set the constant for the first script member of CJK
COMMENT (Chinese/Japanese/Korean) 3 byte weight range
SET constant CJK3BYTESTART to 0xC0
COMMENT Set the constant for the last script member of CJK
COMMENT (Chinese/Japanese/Korean) 3 byte weight range
SET constant CJK3BYTEEND to 0xEF
ENDIF
SET constant FIRST_SCRIPT to LATIN
SET constant MAX_SCRIPTS to 256

//
// Values for CJK Unified Ideographs Extension A range.
// 0x3400 thru 0x4dbf
//
SET constant SCRIPT_MEMBER_EXT_A to 254 // SM for Extension A
SET constant PRIMARY_WEIGHT_EXT_A to 255 // AW for Extension A

//
// Lowest weight values.
// Used to remove trailing DW and CW values.
// Also used to keep illegal values out of sort keys.
//

SET constant MIN_DW to 2
SET constant MIN_CW to 2

//
// Bit mask values.
//

```

```

// Case Weight (CW) - 8 bits:
//   bit 0  => width
//   bit 1,2 => small kana, sei-on
//   bit 3,4 => upper/lower case
//   bit 5  => kana
//   bit 6,7 => contraction
//

SET constant CONTRACTION_8_MASK to 0xc0
SET constant CONTRACTION_7_MASK to 0xc0
SET constant CONTRACTION_6_MASK to 0xc0
SET constant CONTRACTION_5_MASK to 0x80
SET constant CONTRACTION_4_MASK to 0x80
SET constant CONTRACTION_3_MASK to 0x40
SET constant CONTRACTION_2_MASK to 0x40

SET constant CONTRACTION_MASK to 0xc0

ELSE
    COMMENT Otherwise, only 2-character or 3-character contractions
    // are supported.
SET constant CONTRACTION_3_MASK to 0xc0
// Bit-mask to check 2 character contraction or 3 character contraction
SET constant CONTRACTION_2_MASK to 0x80
// Bit-mask to check 2 character contraction
ENDIF

SET constant CASE_UPPER_MASK to 0xe7 // zero out case bits
SET constant CASE_KANA_MASK to 0xdf // zero out kana bit
SET constant CASE_WIDTH_MASK to 0xfe // zero out width bit

//
// Masks to isolate the various bits in the case weight.
//
// NOTE: Bit 2 needs to always equal 1 to avoid getting
//       a byte value of either 0 or 1.
//

SET constant CASE_EXTRA_WEIGHT_MASK to 0xc4
SET constant ISOLATE_KANA to
    (~CASE_KANA_MASK) | CASE_EXTRA_WEIGHT_MASK
SET constant ISOLATE_WIDTH to
    (~CASE_WIDTH_MASK) | CASE_EXTRA_WEIGHT_MASK

//
// Values for East Asia special case primary weights.
//
SET constant PW_REPEAT to 0
SET constant PW_CHO_ON to 1
SET constant MAX_SPECIAL_PW to PW_CHO_ON

//
// Values for weight 5 - East Asia Extra Weights.
//
SET constant WT_FIVE_KANA to 3
SET constant WT_FIVE_REPEAT to 4
SET constant WT_FIVE_CHO_ON to 5

//
// PW Mask for Cho-On:
// Leaves bit 7 on in PW, so it becomes Repeat
// if it follows Kana N.
//
SET constant CHO_ON_PW_MASK to 0x87

//
// Special weight values
//

```

```

SET constant MAP_INVALID_WEIGHT to 0xff

//
// Some Significant Values for Korean Jamo.
// The L, V & T syllables in the 0x1100 Unicode range
// can be composed to characters in the 0xac00 range.
// See The Unicode Standard for details.
//
SET constant NLS_CHAR_FIRST_JAMO          to 0x1100
// Begin Jamo range
SET constant NLS_CHAR_LAST_JAMO           to 0x11f9
// End Jamo range
SET constant NLS_CHAR_FIRST_VOWEL_JAMO    to 0x1160
// First Vowel Jamo
SET constant
    NLS_CHAR_FIRST_TRAILING_JAMO to 0x11a8
// First Trailing Jamo
SET constant
    NLS_JAMO_VOWEL_COUNT to 21
// Number of vowel Jamo (V)
SET constant
    NLS_JAMO_TRAILING_COUNT to 28
// Number of trailing Jamo (L)
SET constant
    NLS_HANGUL_FIRST_COMPOSED to 0xac00
// Begin composed range

//
// Values for Unicode Weight extra weights (e.g. Jamo (old Hangul)).
// The following uses SM for extra UW weights.
//
SET constant ScriptMember_Extra_UnicodeWeight to 255
// Leading Weight / Vowel Weight / Trailing Weight
// according to the current Jamo class.
//
STRUCTURE JamoSortInfoType
(
    // true for an old Hangul sequence
    OldHangulFlag : Boolean

    // true if U+1160 (Hangul Jungseong Filler) used
    FillerUsed : Boolean

    // index to the prior modern Hangul syllable (L)
    LeadingIndex : 8 bit integer

    // index to the prior modern Hangul syllable (V)
    VowelIndex : 8 bit integer

    // index to the prior modern Hangul syllable (T)
    TrailingIndex : 8 bit integer

    // Weight to offset from other old hangul (L)
    LeadingWeight : 8 bit integer

    // Weight to offset from other old hangul (V)
    VowelWeight : 8 bit integer

    // Weight to offset from other old hangul (T)
    TrailingWeight : 8 bit integer
)

// This is the raw data record type from the data table
STRUCTURE JamoStateDataType
(
    // true for an old Hangul sequence
    OldHangulFlag : Boolean

    // index to the prior modern Hangul syllable (L)

```

```

    LeadingIndex : 8 bit integer

    // index to the prior modern Hangul syllable (V)
    VowelIndex : 8 bit integer

    // index to the prior modern Hangul syllable (T)
    TrailingIndex : 8 bit integer

    // weight to distinguish from old Hangul
    ExtraWeight : 8 bit integer

    // number of additional records in this state
    TransitionCount : 8 bit integer

    // Current record in unisort.txt Jamo table:
    JamoRecord : data record

    // SORTTABLES\JAMOSORT\[Character] section
)
COMMENT GetWindowsSortKey
COMMENT
COMMENT On Entry:  SourceString - Unicode String to compute a
COMMENT                               sort key for
COMMENT                               SortLocale   - Locale to determine correct
COMMENT                               linguistic sort
COMMENT                               Flags         - Bit Flag to control behavior
COMMENT                                               of sort key generation.
COMMENT
COMMENT NORM_IGNORENONSPACE   Ignore diacritic weight
COMMENT NORM_IGNORECASE:      Ignore case weight
COMMENT NORM_IGNOREKATYPE:    Ignore Japanese Katakana/Hiraga
COMMENT                               difference
COMMENT NORM_IGNOREWIDTH:     Ignore Chinese/Japanese/Korean
COMMENT                               half-width and full-width difference.
COMMENT
COMMENT On Exit:  SortKey      - Byte array containing the
COMMENT                               computed sort key.
COMMENT

PROCEDURE GetWindowsSortKey(IN SourceString : Unicode String,
                           IN SortLocale : LCID,
                           IN Flags : 32 bit integer,
                           OUT SortKey : BYTE String)

COMMENT Compute flags for sort conditions
COMMENT Based on the case/kana/width flags,
COMMENT turn off bits in case mask when comparing case weight.

SET CaseMask to 0xff

If (NORM_IGNORECASE bit is on in Flags) THEN
    SET CaseMask to CaseMask LOGICAL AND with CASE_UPPER_MASK
ENDIF

If (NORM_IGNOREKATYPE bit is on in Flags) THEN
    SET CaseMask to CaseMask LOGICAL AND with CASE_KANA_MASK
ENDIF

If (NORM_IGNOREWIDTH bit is on in Flags) THEN
    SET CaseMask to CaseMask LOGICAL AND with CASE_WIDTH_MASK
ENDIF

COMMENT Windows 7 and Windows Server 2008 R2 use 3-byte
COMMENT (instead of 2-byte) sequence for Unicode Weights
COMMENT for Private Use Area (PUA) and some Chinese/Japanese/Korean (
COMMENT CJK) script members.
COMMENT Does this sort have a 3-byte Unicode Weight (CJK sorts)?
IF Windows version is Windows 7 and Windows Server 2008 R2 THEN
    COMMENT Check if the locale can have 3-byte Unicode weight

```

```

    SET Is3ByteWeightLocale to CALL Check3ByteWeightLocale(SortLocale)
ENDIF

IF Windows version is Windows Vista, Windows Server 2008, Windows 7, or
Windows Server 2008 R2 THEN
    COMMENT For Windows Vista, Windows Server 2008, Windows 7, and
    COMMENT Windows Server 2008 R2, the algorithm
    COMMENT does not remap the script for Korean locale
    SET IsKoreanLocale to false
ELSE

    IF SortLocale is LCID_KOREAN or
    SortLocale is LCID_KOREAN_UNICODE_SORT THEN
        SET IsKoreanLocale to true
        IF KoreanScriptMap is null THEN
            CALL InitKoreanScriptMap
        ELSE
            SET IsKoreanLocale to false
        ENDIF
    ENDIF

    //
    // Allocate buffer to hold different levels of sort key weights.
    // UnicodeWeights/ExtraWeights/SpecialWeights will be eventually
    // to be collected together, in that order, into the returned
    // Sortkey byte string.
    //
    // Maximum expansion size is 3 times the input size
    //

    // Unicode Weight => 4 word (16 bit) length
    // (extension A and Jamo need extra words)
    SET UnicodeWeights to new empty string of UnicodeWeightType

    SET DiacriticWeights to new empty string of BYTE
    SET CaseWeights to new empty string of BYTE

    // Extra Weight=>4 byte length (4 weights, 1 byte each) FE Special
    SET ExtraWeights to new empty string of ExtraWeightType

    // Special Weight => dword length (2 words each of 16 bits)
    SET SpecialWeights to new empty string of SpecialWeightType

    //
    // Go through the string, code point by code point,
    // testing for contractions and Hungarian special character sequence
    //

    // loop presumes 0 based index for source string
    FOR SourceIndex is 0 to Length(SourceString) -1
        //
        // Get weights
        // CharacterWeight will contain all of the weight information
        // for the character tested.
        //

        SET CharacterWeight to CALL GetCharacterWeights
        WITH (SortLocale, SourceString[SourceIndex])

        SET ScriptMember to CharacterWeight.ScriptMember

        // Special case weights have script members less than
        // MAX_SPECIAL_CASE (11)
        IF ScriptMember is greater than MAX_SPECIAL_CASE THEN

            //
            // No special case on character, but has to check for

```

```

// contraction characters and Hungarian special
// character sequence characters.
//

SET HasHungarianSpecialCharacterSequence to CALL
TestHungarianCharacterSequences
    WITH (SortLocale, SourceString, SourceIndex)

SET Result to CALL GetContractionType WITH (CharacterWeight)

CASE Result OF

    "3-character Contraction":
        COMMENT This is only possible for Windows versions that
        COMMENT are Windows NT 4.0 through Windows Server 2003
        Set ContractionFound to CALL SortkeyContractionHandler
            WITH (SortLocale, SourceString, SourceIndex,
                HasHungarianSpecialCharacterSequence, 3,
                UnicodeWeights, DiacriticWiegths, CaseWeights)
        IF ContractionFound is true THEN
            COMMENT Break out of the case statement
            BREAK
        ENDIF
        IF ContractionFound is true THEN
            COMMENT Break out of the case statement
            BREAK
        ENDIF
        COMMENT If no contraction is found, fall through into additional cases.
        FALLTHROUGH

    "2-character Contraction":
        COMMENT This is only possible for Windows versions that are
        COMMENT Windows NT 4.0 through Windows Server 2003
        Set ContractionFound to CALL SortkeyContractionHandler
            WITH (SortLocale, SourceString, SourceIndex,
                HasHungarianSpecialCharacterSequence, 2,
                UnicodeWeights, DiacriticWiegths, CaseWeights)
        IF ContractionFound is true THEN
            COMMENT Break out of the case statement
            BREAK
        ENDIF
        COMMENT If no contraction is found, fall through into the OTHER case.
        COMMENT Since "3-character contraction" or "2-character contraction"
        COMMENT are the only two possible values for
        COMMENT Windows NT 4.0 through Windows Server 2003, all calls to
        COMMENT SortkeyContractionHandler will return false.
        COMMENT So, the fallthrough will go directly to the OTHERS section
        FALLTHROUGH

    "6-character contraction, 7-character contraction, or 8-character
contraction":
        Set ContractionFound to CALL SortkeyContractionHandler
            WITH (SortLocale, SourceString, SourceIndex,
                HasHungarianSpecialCharacterSequence, 8,
                UnicodeWeights, DiacriticWiegths, CaseWeights)
        IF ContractionFound is true THEN
            COMMENT Break out of the case statement
            BREAK
        ELSE
            Set ContractionFound to CALL SortkeyContractionHandler
                WITH (SortLocale, SourceString, SourceIndex,
                    HasHungarianSpecialCharacterSequence, 7,
                    UnicodeWeights, DiacriticWiegths, CaseWeights)
        ENDIF
        IF ContractionFound is true THEN
            COMMENT Break out of the case statement
            BREAK
        ELSE
            Set ContractionFound to CALL SortkeyContractionHandler

```

```

        WITH (SortLocale, SourceString, SourceIndex,
              HasHungarianSpecialCharacterSequence, 6,
              UnicodeWeights, DiacriticWeights, CaseWeights)
    ENDIF
    IF ContractionFound is true THEN
        COMMENT Break out of the case statement
        BREAK
    ENDIF
    COMMENT If no contraction is found, fall through into additional cases.
    FALLTHROUGH

    "4-character contraction or 5-character contraction":
    Set ContractionFound to CALL SortkeyContractionHandler
    WITH (SortLocale, SourceString, SourceIndex,
          HasHungarianSpecialCharacterSequence, 5,
          UnicodeWeights, DiacriticWeights, CaseWeights)
    IF ContractionFound is true THEN
        COMMENT Break out of the case statement
        BREAK
    ELSE
        Set ContractionFound to CALL SortkeyContractionHandler
        WITH (SortLocale, SourceString, SourceIndex,
              HasHungarianSpecialCharacterSequence, 4,
              UnicodeWeights, DiacriticWeights, CaseWeights)
    ENDIF
    IF ContractionFound is true THEN
        COMMENT Break out of the case statement
        BREAK
    ENDIF
    COMMENT If no contraction is found, fall through into additional cases.
    FALLTHROUGH

    "2-character contraction or 3-character contraction":
    Set ContractionFound to CALL SortkeyContractionHandler
    WITH (SortLocale, SourceString, SourceIndex,
          HasHungarianSpecialCharacterSequence, 3,
          UnicodeWeights, DiacriticWeights, CaseWeights)
    IF ContractionFound is true THEN
        COMMENT Break out of the case statement
        BREAK
    ELSE
        Set ContractionFound to CALL SortkeyContractionHandler
        WITH (SortLocale, SourceString, SourceIndex,
              HasHungarianSpecialCharacterSequence, 2,
              UnicodeWeights, DiacriticWeights, CaseWeights)
    ENDIF
    IF ContractionFound is true THEN
        COMMENT Break out of the case statement
        BREAK
    ENDIF
    COMMENT If no contraction is found, fall through into additional cases.
    FALLTHROUGH

    OTHERS :
    IF Windows version is greater than Windows Server 2008 R2 or Windows 7
    THEN
        COMMENT In Windows Server 2008 R2 or Windows 7,
        COMMENT Private Use Area (PUA) code points
        COMMENT and some CJK (Chinese/Japanese/Korean) sorts
        COMMENT might need 3 byte weights
        COMMENT Store normal Unicode weight first. Note that there is no
        COMMENT adjustment of Korean weight anymore.
        SET UnicodeWeight to
            CorrectUnicodeWeight(CharacterWeight, FALSE)
        COMMENT Assume 3-byte Unicode Weight is not used first.
        COMMENT The algorithm will check this later.
        SET UnicodeWeight.ThirdByteWeight to 0
    
```

```

        IF (ScriptMember is equal to or greater than PUA3BYTESTART)
            AND
            (ScriptMember is less than or equal to PUA3BYTEEND) THEN
            SET IsScriptMemberPUA3ByteWeight to true
        ELSE
            SET IsScriptMemberPUA3ByteWeight to false
        ENDIF

        IF (ScriptMember is equal to or greater than CJK3BYTESTART) AND
            (ScriptMember is less than or equal to CJK3BYTEEND) THEN
            SET IsScriptMemberCJK3ByteWeight to true
        ELSE
            SET IsScriptMemberCJK3ByteWeight to false
        ENDIF
        IF (IsScriptMemberPUA3ByteWeight is true) OR
            (Is3ByteWeightLocale AND
            IsScriptMemberCJK3ByteWeight is true) THEN
            COMMENT PUA code points and some CJK sorts need 3 byte weights
            SET UnicodeWeight.ThirdByteWeight to
CharacterWeight.DiacriticWeight
        ELSE
            COMMENT Normal Diacritic Weight
            APPEND CharacterWeight.DiacriticWeight to DiacriticWeights as a
BYTE
        ENDIF
        APPEND UnicodeWeight to UnicodeWeights

        SET CaseWeight to GetCaseWeight(CharacterWeight)
        APPEND CharacterWeight.CaseWeight to CaseWeights as a BYTE

    ELSE

        SET UnicodeWeight to
        CorrectUnicodeWeight(CharacterWeight, IsKoreanLocale)
        APPEND UnicodeWeight to UnicodeWeights
        APPEND CharacterWeight.DiacriticWeight to DiacriticWeights
        as a BYTE
        SET CaseWeight to GetCaseWeight(CharacterWeight)
        APPEND CharacterWeight.CaseWeight to CaseWeights as a BYTE
    ENDIF
ENDCASE
ELSE
    CALL SpecialCaseHandler WITH (SourceString, SourceIndex,
        UnicodeWeights, ExtraWeights, SpecialWeights,
        SortLocale, IsKoreanLocale)
ENDIF
ENDFOR

//
// Store the Unicode Weights in the destination buffer.
//
FOR each UnicodeWeight in UnicodeWeights
    //
    // Copy Unicode weight to destination buffer.
    //
    APPEND UnicodeWeight.ScriptMember to SortKey as a BYTE
    APPEND UnicodeWeight.PrimaryWeight to SortKey as a BYTE
    IF Windows version is greater than Windows Server 2008 R2 or Windows 7 THEN
        IF UnicodeWeight.ThirdByteWeight is not 0 THEN
            COMMENT When 3-byte Unicode Weight is used, append the additional
            COMMENT BYTE into SortKey
            APPEND UnicodeWeight.ThirdByteWeight to SortKey as a BYTE
        ENDIF
    ENDIF
ENDIF
ENDFOR

```

```

//
// Copy Separator to destination buffer.
//
APPEND SORTKEY_SEPARATOR to SortKey as a BYTE

//
// Store Diacritic Weights in the destination buffer.
//
IF (NORM_IGNORENONSPACE bit is not turned on in Flags) THEN
  IF (IsReverseDW is TRUE) THEN
    //
    // Reverse diacritics:
    // - remove diacritics from left to right.
    // - store diacritics from right to left.
    //
    FOR each DiacriticWeight in
      DiacriticWeights in the "first in first out" order
      IF DiacriticWeight <= MIN_DW THEN
        REMOVE DiacriticWeight from DiacriticWeights
      ELSE
        BREAK from the current FOR loop
      ENDIF
    ENDFOR

    FOR each DiacriticWeight in
      DiacriticWeights in the "last in first out" order
      //
      // Copy Unicode weight to destination buffer.
      //
      APPEND DiacriticWeight to SortKey as a BYTE
    ENDFOR
  ELSE
    //
    // Regular diacritics:
    // - remove diacritics from right to left.
    // - store diacritics from left to right.
    //
    FOR each DiacriticWeight in
      DiacriticWeights in the "last in first out" order
      IF DiacriticWeight <= MIN_DW THEN
        REMOVE DiacriticWeight from DiacriticWeights
      ELSE
        BREAK from the current FOR loop
      ENDIF
    ENDFOR

    FOR each DiacriticWeight in
      DiacriticWeights in the order of "first in first out"
      //
      // Copy Unicode weight to destination buffer.
      //
      APPEND DiacriticWeight to SortKey as a BYTE
    ENDFOR
  ENDIF
ENDIF

//
// Copy Separator to destination buffer.
//
APPEND SORTKEY_SEPARATOR to SortKey as a BYTE

//
// Store case Weights
//
// - Eliminate minimum CW.
// - Copy case weights to destination buffer.
//
IF (NORM_IGNORECASE bit is not turned on in Flags
  OR NORM_IGNOREWIDTH bit is not turned on in Flags) THEN
  FOR each CaseWeight in CaseWeights

```

```

        in the "last in first out" order
        IF CaseWeight <= MIN_CW THEN
            REMOVE CaseWeight from CaseWeights
        ELSE
            BREAK from the current FOR loop
        ENDIF
    ENDFOR

    FOR each CaseWeight in CaseWeights
        //
        // Copy Unicode weight to destination buffer.
        //
        APPEND CaseWeight to SortKey as a BYTE
    ENDFOR
ENDIF

//
// Copy Separator to destination buffer.
//
APPEND SORTKEY_SEPARATOR to SortKey as a BYTE

//
// Store the Extra Weights in the destination buffer for
// EAST ASIA Special.
//
// - Eliminate unnecessary XW.
// - Copy extra weights to destination buffer.
//
IF Length(ExtraWeights) is greater than 0 THEN
    IF (NORM_IGNORENONSPACE bit is turned on in Flag) THEN
        APPEND 0xff to SortKey as a BYTE
        APPEND 0x02 to SortKey as a BYTE
    ENDIF

    // Append W6 group to SortKey
    // Trim unused values from the end of the string
    SET EndExtraWeight to Length(ExtraWeights) - 1

    WHILE EndExtraWeight greater than 0 and
        ExtraWeightSeparator[EndExtraWeight].W6 == 0xe4
        DECREMENT EndExtraWeight
    ENDWHILE

    SET ExtraWeightIndex to 0
    WHILE ExtraWeightIndex is less than or equal to EndExtraWeight
        APPEND ExtraWeightSeparator[ExtraWeightIndex].W6
        to SortKey as a BYTE
        INCREMENT ExtraWeightIndex
    ENDWHILE

    // Append W6 separator
    APPEND 0xff to SortKey as a BYTE

    // Append W7 group to SortKey
    // Trim unused values from the end of the string
    SET EndExtraWeight to Length(ExtraWeights) - 1
    WHILE EndExtraWeight greater than 0 and
        ExtraWeightSeparator[EndExtraWeight].W7 == 0xe4
        DECREMENT EndExtraWeight
    ENDWHILE

    SET ExtraWeightIndex to 0
    WHILE ExtraWeightIndex is less than or equal to EndExtraWeight
        APPEND ExtraWeightSeparator[ExtraWeightIndex].W7 to SortKey
        INCREMENT ExtraWeightIndex
    ENDWHILE

    // Append W7 separator
    APPEND 0xff to SortKey as a BYTE

```

```

ENDIF

//
// Copy Separator to destination buffer.
//
APPEND SORTKEY_SEPARATOR to SortKey as a BYTE

//
// Store the Special Weights in the destination buffer.
//
// - Copy special weights to destination buffer.
//
FOR each SpecialWeight in SpecialWeights
    // High byte (most significant)
    SET Byte1 to SpecialWeight.Position >> 8
    // Low byte (least significant)
    SET Byte2 to SpecialWeight.Position & 0xff
    APPEND Byte1 to SortKey as a BYTE
    APPEND Byte2 to SortKey as a BYTE
    APPEND SpecialWeight.Script to SortKey as a BYTE
    APPEND SpecialWeight.Weight to SortKey as a BYTE
ENDFOR

//
// Copy terminator to destination buffer.
//
APPEND SORTKEY_TERMINATOR to SortKey

RETURN SortKey

```

<5> [Section 3.1.5.2.16](#): The following MapOldHangulSortKey algorithm is only used in Windows NT, Windows 2000, Windows XP, Windows Server 2003, Windows Vista, Windows Server 2008, Windows 7, and Windows Server 2008 R2.

```

COMMENT MapOldHangulSortKey
COMMENT
COMMENT On Entry:  SourceString - Unicode String to test
COMMENT              SourceIndex - Index of leading Jamo to start
COMMENT                      from
COMMENT              SortLocale - Locale to use for linguistic
COMMENT                      sort data
COMMENT              UnicodeWeights - String to store any Unicode
COMMENT                      weight found
COMMENT                      for this character(s)
COMMENT
COMMENT On Exit:   CharactersRead - Number of old Hangul found
COMMENT              UnicodeWeights - Any Unicode weights found are
COMMENT                      appended
COMMENT

PROCEDURE MapOldHangulSortKey(IN SourceString : Unicode String,
                              IN SourceIndex : 32 bit integer,
                              IN SortLocale : LCID,
                              IN OUTUnicodeWeights : String of UnicodeWeightType,
                              IN IsKoreanLocale : Boolean,
                              OUT CharactersRead : 32 bit integer)

SET CurrentIndex to SourceIndex
SET JamoSortInfo to empty JamoSortInfoType

// Get any Old Hangul Leading Jamo composition for our Leading Jamo
SET JamoClass to CALL GetJamoComposition WITH (SourceString,
                                                SourceIndex, "Leading Jamo Class", JamoSortInfo)

IF JamoClass is equal to "Vowel Jamo Class" THEN
    // A Vowel Jamo, try to find an

```

```

        // Old Hangul Vowel Jamo composition.
        SET JamoClass to CALL GetJamoComposition WITH (SourceString,
            SourceIndex, "Vowel Jamo Class", JamoSortInfo)
    ENDIF

    IF JamoClass is equal to "Trailing Jamo Class" THEN
        // A Trailing Jamo, try to find an
        // Old Hangul Trailing Jamo composition.
        SET JamoClass CALL GetJamoComposition WITH (SourceString,
            SourceIndex, "Trailing Jamo Class", JamoSortInfo)
    ENDIF

    // A valid leading and vowel sequence and this is
    // old Hangul...
    IF JamoSortInfo.OldHangulFlag is true THEN

        // Compute the modern hangul syllable prior to this composition
        // Users formula from Unicode 3.0 Section 3.11 p54
        // "Hangul Syllable Composition"
        // This converts a U+11.. sequence to a U+AC00 character

        SET ModernHangul to (JamoSortInfo.LeadingIndex *
            NLS_JAMO_VOWELCOUNT + JamoSortInfo.VowelIndex) *
            NLS_JAMO_TRAILING_COUNT + JamoSortInfo.TrailingIndex +
            NLS_HANGUL_FIRST_SYLLABLE

        IF JamoSortInfo.FillerUsed is true THEN
            // If the filler is used, sort before the modern Hangul,
            // instead of after
            DECREMENT ModernHangul

            // If falling off the modern Hangul syllable block...
            IF ModernHangul is less than NLS_HANGUL_FIRST_SYLLABLE THEN
                // Sort after the previous character
                // (Circled Hangul Kiyeok A)
                SET ModernHangul to 0x326e
            ENDIF

            // Shift the leading weight past any old Hangul
            // that sorts after this modern Hangul
            SET JamoSortInfo.LeadingWeight to
                JamoSortInfo.LeadingWeight + 0x80
        ENDIF

        // Store the weights
        SET CharacterWeight to CALL GetCharacterWeights WITH (ModernHangul)
        SET UnicodeWeight to CALL CorrectUnicodeWeight
            WITH (CharacterWeight, IsKoreanLocale)
        APPEND UnicodeWeight to UnicodeWeights

        // Add additional weights
        SET UnicodeWeight to CALL MakeUnicodeWeight WITH
            (ScriptMember_Extra_UnicodeWeight,
                JamoSortInfo.LeadingWeight, false)
        APPEND UnicodeWeight to UnicodeWeights

        SET UnicodeWeight to CALL MakeUnicodeWeight WITH
            (ScriptMember_Extra_UnicodeWeight,
                JamoSortInfo.VowelWeight, false)

        APPEND UnicodeWeight to UnicodeWeights
        SET UnicodeWeight to CALL MakeUnicodeWeight WITH
            (ScriptMember_Extra_UnicodeWeight,
                JamoSortInfo.TrailingWeight, false)

        APPEND UnicodeWeight to UnicodeWeights

        // Return the characters consumed
        SET CharactersRead to CurrentIndex - SourceIndex

```

```

        RETURN CharactersRead
    ENDIF

    // Otherwise it isn't a valid old Hangul composition
    // and don't do anything with it

    SET CharactersRead to 0
    RETURN CharactersRead

```

<6> [Section 3.1.5.2.17](#): The GetJamoComposition algorithm is only used in Windows NT, Windows 2000, Windows XP, Windows Server 2003, Windows Vista, Windows Server 2008, Windows 7, and Windows Server 2008 R2.

<7> [Section 3.1.5.2.18](#): The following GetJamoStateData algorithm is only used in Windows NT, Windows 2000, Windows XP, Windows Server 2003, Windows Vista, Windows Server 2008, Windows 7, and Windows Server 2008 R2.

```

COMMENT GetJamoStateData
COMMENT
COMMENT On Entry:  Character      - Unicode Character to get Jamo
COMMENT                               information for
COMMENT
COMMENT On Exit:   JamoStateData - Jamo state information from
COMMENT                               the data file
COMMENT
COMMENT Jamo State information looks like this in the database:
COMMENT
COMMENT SORTTABLES
COMMENT     ...
COMMENT     JAMOSORT395
COMMENT     ...
COMMENT     0x11724
COMMENT     0x1172 0x00 0x00 0x11 0x00 0x38 0x03; U+1172
COMMENT     0x1161 0x01 0x00 0x00 0x00 0x00 0x01; U+1172,1161
COMMENT     0x1175 0x01 0x00 0x11 0x1b 0x3a 0x00; U+1172,1161,1175
COMMENT     0x1169 0x01 0x00 0x11 0x1b 0x3f 0x00; U+1172,1169

PROCEDURE GetJamoStateData (IN Character : Unicode Character,
                           OUT JamoStateData : JamoStateDataType)

// Get the Jamo section for this character.
// If Character was 0x1172, this would access the following section:
// 0x11724
//   0x1172 0x00 0x00 0x11 0x00 0x38 0x03 ; U+1172           record 0
//   0x1161 0x01 0x00 0x00 0x00 0x00 0x01 ; U+1172,1161      record 1
//   0x1175 0x01 0x00 0x11 0x1b 0x3a 0x00 ; U+1172,1161,1175 record 2
//   0x1169 0x01 0x00 0x11 0x1b 0x3f 0x00 ; U+1172,1169      record 3
//   |       |       |       |       |       |       |
//   Field 1 2       3       4       5       6       7       Comment
//
OPEN SECTION JamoSection
    where name is SORTTABLES\JAMOSORT\[Character] from unisort.txt

// Now open the first record
SELECT RECORD JamoRecord FROM JamoSection WHERE record index is 0

// Now gather the information from that record.
SET JamoStateData.OldHangulFlag to JamoRecord.Field2
SET JamoStateData.LeadIndex to JamoRecord.Field3
SET JamoStateData.VowelIndex to JamoRecord.Field4
SET JamoStateData.TrailingIndex to JamoRecord.Field5
SET JamoStateData.ExtraWeight to JamoRecord.Field6
SET JamoStateData.TransitionCount to JamoRecord.Field7

// Remember the record
SET JamoStateData.DataRecord to JamoRecord

```

```
RETURN JamoStateData
```

<8> [Section 3.1.5.2.19](#): The FindNewJamoState algorithm is only used in Windows NT, Windows 2000, Windows XP, Windows Server 2003, Windows Vista, Windows Server 2008, Windows 7, and Windows Server 2008 R2.

<9> [Section 3.1.5.2.20](#): The following UpdateJamoSortInfo algorithm is only used in Windows NT, Windows 2000, Windows XP, Windows Server 2003, Windows Vista, Windows Server 2008, Windows 7, and Windows Server 2008 R2.

```
COMMENT UpdateJamoSortInfo
COMMENT
COMMENT On Entry:  JamoClass      - The current Jamo Class
COMMENT              JamoStateData - Information about the new
COMMENT                      character state
COMMENT              JamoSortInfo  - Information about the character
COMMENT                      state
COMMENT On Exit:   JamoSortInfo  - Updated with information about
COMMENT                      the new state based on JamoClass
COMMENT                      and JamoSortData
COMMENT

PROCEDURE UpdateJamoSortInfo(IN JamoClass : enumeration,
                             IN JamoStateData : JamoStateDataType,
                             INOUT JamoSortInfo : JamoSortInfoType)

// Record if this is a Jamo unique to old Hangul
SET JamoSortInfo.OldHangulFlag to
    JamoSortInfo.OldHangulFlag | JamoStateData.OldHangulFlag

// Update the indices if the new ones are higher than the current
// ones.
IF JamoStateData.LeadingIndex
    is greater than JamoSortInfo.LeadingIndex THEN
    SET JamoSortInfo.LeadingIndex to JamoStateData.LeadingIndex;
ENDIF

IF JamoStateData.VowelIndex
    is greater than JamoSortInfo.VowelIndex THEN
    SET JamoSortInfo.VowelIndex to JamoStateData.VowelIndex;
ENDIF

IF JamoStateData.TrailingIndex
    is greater than JamoSortInfo.TrailingIndex THEN
    SET JamoSortInfo.TrailingIndex to JamoStateData.TrailingIndex;
ENDIF

// Update the extra weights according to the current Jamo class.
CASE JamoClass OF
    "Leading Jamo Class":
        IF JamoStateData.ExtraWeight
            is greater than JamoSortInfo.LeadingWeight THEN
            SET JamoSortInfo.LeadingWeight to JamoStateData.ExtraWeight
        ENDIF

    "Vowel Jamo Class":
        IF JamoStateData.ExtraWeight
            is greater than JamoSortInfo.VowelWeight THEN
            SET JamoSortInfo.VowelWeight to JamoStateData.ExtraWeight
        ENDIF

    "Trailing Jamo Class":
        IF JamoStateData.ExtraWeight
            is greater than JamoSortInfo.TrailingWeight THEN
```

```

        SET JamoSortInfo.TrailingWeight to JamoStateData.ExtraWeight
    ENDIF
ENDCASE

RETURN JamoSortInfo

```

[<10> Section 3.1.5.2.21](#): The IsJamo algorithm is only used in Windows NT, Windows 2000, Windows XP, Windows Server 2003, Windows Vista, Windows Server 2008, Windows 7, and Windows Server 2008 R2.

[<11> Section 3.1.5.2.22](#): The IsCombiningJamo algorithm is not used in Windows NT, Windows 2000, Windows XP, Windows Server 2003, Windows Vista, Windows Server 2008, Windows 7, and Windows Server 2008 R2.

[<12> Section 3.1.5.2.23](#): The following IsJamoLeading algorithm is only used in Windows NT, Windows 2000, Windows XP, Windows Server 2003, Windows Vista, Windows Server 2008, Windows 7, and Windows Server 2008 R2.

```

COMMENT IsJamoLeading
COMMENT
COMMENT On Entry:  SourceCharacter - Unicode Character to test
COMMENT
COMMENT On Exit:   Result           - true if SourceCharacter is a
COMMENT                                     leading Jamo
COMMENT
COMMENT NOTE: Only call this if the character is known to be a Jamo
COMMENT          syllable. This function only helps distinguish between
COMMENT          the different types of Jamo, so only call it if
COMMENT          IsJamo() has returned true.
COMMENT

PROCEDURE IsJamoLeading(IN SourceCharacter : Unicode Character,
                      OUT Result: boolean)

IF SourceCharacter is less than NLS_CHAR_FIRST_VOWEL_JAMO THEN
    SET Result to true
ELSE
    SET Result to false
ENDIF

RETURN Result

```

[<13> Section 3.1.5.2.24](#): The IsJamoVowel algorithm is not applicable to Windows NT, Windows 2000, Windows XP, Windows Server 2003, Windows Vista, Windows Server 2008, Windows 7, and Windows Server 2008 R2.

[<14> Section 3.1.5.2.25](#): The following IsJamoTrailing algorithm is only used in Windows NT, Windows 2000, Windows XP, Windows Server 2003, Windows Vista, Windows Server 2008, Windows 7, and Windows Server 2008 R2.

```

COMMENT IsJamoTrailing
COMMENT
COMMENT On Entry:  SourceCharacter - Unicode Character to test
COMMENT
COMMENT On Exit:   Result           - true if this is a trailing Jamo
COMMENT
COMMENT NOTE: Only call this if the character is known to be a Jamo
COMMENT          syllable. This function only helps distinguish between
COMMENT          the different types of Jamo, so only call it if
COMMENT          IsJamo() has returned true.
COMMENT

```

```

PROCEDURE IsJamoTrailing(IN SourceCharacter : Unicode Character,
                        OUT Result: boolean)

IF SourceCharacter is greater than
  or equal to NLS_CHAR_FIRST_VOWEL_JAMO THEN
  SET Result to true
ELSE
  SET Result to false
ENDIF

RETURN Result

```

[<15> Section 3.1.5.4](#): The IdnToNameprepUnicode, IdnToAscii, and IdnToUnicode algorithms are not applicable to Windows NT, Windows 2000, Windows XP, or Windows Server 2003. These algorithms follow the **IDNA2003** standards for Windows Vista, Windows Server 2008, Windows 7, and Windows Server 2008 R2 operating system. Otherwise, the algorithms follow the **IDNA2008+UTS46** standards.

[<16> Section 3.1.5.4.6](#): This version is not used in Windows NT, Windows 2000, Windows XP, Windows Server 2003, Windows Vista, Windows Server 2008, Windows 7, and Windows Server 2008 R2.

[<17> Section 3.1.5.4.7](#): This version is used in Windows Vista, Windows Server 2008, Windows 7, and Windows Server 2008 R2

7 Change Tracking

This section identifies changes that were made to this document since the last release. Changes are classified as Major, Minor, or None.

The revision class **Major** means that the technical content in the document was significantly revised. Major changes affect protocol interoperability or implementation. Examples of major changes are:

- A document revision that incorporates changes to interoperability requirements.
- A document revision that captures changes to protocol functionality.

The revision class **Minor** means that the meaning of the technical content was clarified. Minor changes do not affect protocol interoperability or implementation. Examples of minor changes are updates to clarify ambiguity at the sentence, paragraph, or table level.

The revision class **None** means that no new technical changes were introduced. Minor editorial and formatting changes may have been made, but the relevant technical content is identical to the last released version.

The changes made to this document are listed in the following table. For more information, please contact dochelp@microsoft.com.

Section	Description	Revision class
3.1.5.2.3.1 Windows Sorting Weight Table	Updated for Windows 11 v24H2 and Windows Server 2025. Added Windows Server 2022 and Windows 11 to table in PBN.	Major
3.1.5.2.4 GetWindowsSortKey Pseudocode	Updated for Windows 11 v24H2 and Windows Server 2025. Updated GetWindowsSortKey algorithm and added sort key algorithm to PBN.	Major
6 Appendix A: Product Behavior	Added Windows Server 2025 to the product applicability list.	Major

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